

Analysis of Household Surveys

Session notes

Ethan Ligon

September 17, 2026

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1 Design, Sampling, and Collection

1.1 Reading

- Deaton ch. 1 — the whole chapter
- Deaton §1.4 on survey design and §2.2 on what to do about it
- Skim the GLSS7 *Report* methodology section

Deaton is open access (CC BY 3.0 IGO). On the hub it is already in your `reading/` folder; otherwise PDF.

The GLSS7 report is published by the Ghana Statistical Service and is not ours to redistribute: GSS catalogue 97.

1.2 Setup

```
%xmode Plain
```

```
# The library audits its own corpus on first read and reports what it finds  
# — implausible quantities, NaN index keys, a column that is wholly null  
# one wave — at multi-paragraph length. Those reports are a work queue for  
# whoever maintains the data, not something the room can act on, and they block  
# the output they are attached to. Silenced here by their own  
# "Set LSMS_..._STRICT=1" signature, which is precise: every other warning,  
# pandas deprecations included, still shows. Delete these two lines to read  
# them.
```

```
import warnings
```

```
warnings.filterwarnings("ignore",  
                        message=r"(?s).*Set_LSMS_[A-Z_]+=1_to_make_this_fatal")
```

```
import matplotlib as mpl
```

```
mpl.rcParams.update({'figure.dpi': 110, 'font.size': 11,  
                    'lines.linewidth': 1.8})
```

```
import lsms_library as ll
```

```
import pandas as pd
```

1.3 Brief description of LSMS Surveys

A timeline of the surveys `lsms_library` knows about: one row per country, one bar per survey wave, so we can see overall temporal coverage — 37

countries, 113 waves, 1985 to the present.

1.4 Topical coverage of LSMS Surveys

You can check current coverage yourself:

```
cov = ll.coverage()
cov
```

sane means the library builds that feature for that wave and the result passes its checks; **absent** and **undeclared** mean it is not there to build.

Summarizing:

```
print("countries: %d | features: %d | waves: %d"
      % (cov.country.nunique(), cov.feature.nunique(),
         cov[cov.wave != ""].wave.nunique()))
cov.tier.value_counts()
```

So what is available for a country you care about:

```
cov.head(10)
```

1.5 LSMS_Library is a work in progress

Open source; see https://github.com/ligon/lms_library. The idea is to provide a software *abstraction layer* to interface with LSMS-style surveys, providing a set of common labels, tables, and variables.

Note that this is **not** "harmonization"; we preserve all the detail of the individual surveys. Aggregation across surveys where necessary is left to the analyst.

If you find a problem, something's missing, or have a feature request: https://github.com/ligon/lms_library/issues. I'd really appreciate your help!

1.6 Explore Example Country

What countries are available?

```
ll.countries()
```

I'll use Uganda as my example; it's one of the best developed countries in `lms_library`. But I encourage you to play with others; just change the string 'Uganda' below.

```
eg = ll.Country('Uganda')
```

```
eg.waves
```

What different kinds of features/data are available for this country?

```
eg.data_scheme
```

What does the sample look like?

```
eg.sample()
```

Some actual data:

```
eg.household_roster().sort_index()
```

1.7 Survey Design

Any household survey has:

Population What is the *population* that is supposed to be represented?

Sample A collection of selected households

Sample Frame What is the practical information we have on the population that we can use to construct the sample?

Sample Weights Proportional to the *ex ante* probability that a given sample household was selected for inclusion in the sample.

Sample Data Actual information collected from actual households in the sample

1.8 Survey Design Example

Consider the Ghana Living Standards Survey (GhanaLSS), and specifically the most recent publicly released wave 7 (2016–17).

Sample Frame A list of private dwellings from the 2010 census. What does this exclude?

Sample Weights Proportional to the inverse of the *ex ante* probability that a given sample household was selected for inclusion in the sample. The weights in `lsms_library` are normalized to have a mean of one, but sometimes instead they're normalized to sum to the estimated size of the population.

1.9 Survey Design Example

Consider the Ghana Living Standards Survey (GhanaLSS), and specifically the most recent publicly released wave 7 (2016–17).

Population

```
glss = ll.Country('GhanaLSS')
WAVE = '2016-17'

glss.population[WAVE].population_statement
```

Sample

```
glss7 = glss[WAVE]
glss7.sample()
```

Sample Frame A list of private dwellings from the 2010 census. What does this exclude?

Sample Weights Proportional to the inverse of the *ex ante* probability that a given sample household was selected for inclusion in the sample. The weights in `lsms_library` are normalized to have a mean of one, but sometimes instead they're normalized to sum to the estimated size of the population.

Sample Data Actual information collected from actual households in the sample. For example, here's some information on the individuals within each household in the GhanaLSS 2016-17 data:

```
glss7.household_roster()
```

There are many different similar tables available; to see them

```
glss.features
```

1.10 Choosing a Population

In designing a population survey, one has to first choose a population. We'll be focused on *household* surveys within a particular country.

That sounds like a pretty good start: for example, all households in Ghana. But almost immediately edge cases turn up.

- First, what's a household? Usual definition: a set of people who "share a cooking pot"

1.11 Who is outside the household population?

What about people in Ghana who don't live in private households?

- Convicts who live in prisons
- Students who live in dormitories
- Nuns who live in monasteries
- **Typical solution:** "private households" (so we're excluding the above)

1.12 Choosing a Survey Frame

If we've settled on "private households", we have to figure out a way to actually somehow enumerate all the private households in a country so we can make a random selection.

- Possibilities
 - Use a census (the usual LSMS approach)
 - Use remote sensing and machine learning to try to identify private dwellings (Togo a recent example)
- Problems
 - Census is usually out of date
 - Remote sensing is highly error prone
- Leads to demand for ground-truthing

1.13 Census approach to enumerating households

Typical approach:

1. Divide entire country into "Enumeration Areas" (EAs).
2. Send a team to each EA; have them identify/map all residential structures (separate dwellings, apartment blocks, compounds).
3. Find someone who lives in each structure; if it's a private dwelling, list all who live there.

1.14 Census approach to enumerating households

Step 3 in full:

Find someone who lives in each structure; if it's a private dwelling, list all who live there. If it's an apartment block, get access, and list all who live in each apartment. If it's a compound with multiple dwellings: list regular occupants of each. Are cooking duties shared across dwellings? If so it's a single household; otherwise multiple HHs.

1.15 Two-Stage Cluster Based Sampling

At the time a census is conducted, it's usually the most reliable "ground-truth"; it provides a list of private households, you could just randomly select. But:

- Census out of date almost immediately; new construction, people move, etc.

1.16 Sampling in two stages

1. Take a random sample of EAs.
2. In those EAs, send a team to update the household listing.
3. Randomly sample households from each updated list.

The first-stage draw selects areas; the second-stage draw selects households. Updating only the selected areas keeps the listing feasible.

1.17 Example: Two-Stage Sampling Weights

Guinea-Bissau provides a fairly simple example. It stratifies by region (see below); within a region it uses a simple two-stage sample. Consider the region Bafata; here is a map of the sample EAs chosen in the first stage.

```
ll.coordinate_map('Guinea-Bissau', where={'Region': 'bafata'})
```

1.18 Sampling Weights

We can always compute statistics (mean, variance, etc.) that are valid for the *sample*; but usually we'd like to use the sample to say something about the *population*. For this we need to know the probability π_i that a particular sample household i was included in the sample *ex ante*.

Once we know these we can compute statistics from a sample of size N .

For the sample:

Mean $\bar{x} = \frac{1}{N} \sum_{i \in \text{Sample}} x_i$

Variance $\text{Var}(x) = \frac{1}{N} \sum_{i \in \text{Sample}} x_i^2 - \bar{x}^2$

1.19 Sampling Weights

For the population

Let $\hat{X} = \frac{\sum_{i \in \text{Sample}} x_i / \pi_i}{\sum_{i \in \text{Sample}} 1 / \pi_i}$ (this is the Hájek estimator).

Mean $\bar{X} = \mathbb{E} \left[\hat{X} \right] + O(1/N)$

Variance $\mathbb{E}(X - \bar{X})^2 = S^2 = \mathbb{E} \left[\frac{\sum_{i \in \text{Sample}} (x_i - \hat{X})^2 / \pi_i}{\sum_{i \in \text{Sample}} 1 / \pi_i} \right] + O(1/N)$

Takeaway: To draw inferences about the population we need to know sampling probabilities π_i . *It is not enough to just have the data!*

1.20 Sub-Populations

Conducting a large-scale household survey is expensive! Sometimes you'd like to draw inferences not just about the original population (e.g., all the private households in Ghana), but also for subpopulations: e.g.,

- Particular regions
- Rural vs. Urban

The GLSS: Later rounds of the GhanaLSS are designed to do *both*; for example, to make population statements about the urban population of the Ashanti region.

1.21 Stratification

The most statistically efficient way to do inference for sub-populations is via **stratification**: Partition the overall population into different parts, and (effectively) treat each one as it's own sampling problem.

- Let N_s be the size of the sample for stratum s , and n_s be the size of the population in the stratum.

1.22 Strata Weights

If $N_s/n_s \neq N/n$ (the sample drawn in stratum s isn't proportional to the share of s in the population) then we've violated the maxim that every household in the overall population has an equal chance of being selected. To correct this, construct strata weights

$$v_s = \frac{N_s/n_s}{N/n}$$

to correct for the fact that a household in stratum s has a probability of being selected into the sample which must be adjusted by the factor v_s .

1.23 Two-Stage Sampling Weights

In a Two-Stage Clustered sampling scheme, the probability of a particular household i who lives in EA c_i being included in the sample depends on two other probabilities:

- The probability of EA (cluster) c_i being selected in the first stage, say p_{c_i} ;
- *Conditional* on c_i being selected, the probability of i being selected from all the other households in the cluster, say r_i .

So, in a two stage clustering scheme:

$$\pi_i = p_{c_i} r_i$$

1.24 Example: Two-Stage Sampling Weights

Back to our Bafata example.

```
ll.coordinate_map('Guinea-Bissau', size='weight', where={'Region': 'bafata'})

s = ll.Country("Guinea-Bissau").sample().reset_index()
b = s[s["strata"] == "Bafata"]

cl = b.groupby("v").agg(w=("weight", "first"),
                        urban=("Rural", "first"),
                        take=("weight", "size"))
cl["w"] = cl["w"].astype(float)

print("clusters_": len(cl))
```

```
print("households:", len(b))
print("take_": dict(cl["take"].value_counts()))
```

Fifty clusters, 598 households, and forty-eight of the fifty contain exactly twelve households — the two that hold eleven are non-response, not design. This is the textbook object: draw clusters, take a fixed number from each.

```
cv = b.groupby("v")["weight"].std().fillna(0) / b.groupby("v")["weight"].mean()
print("clusters_whose_weight_varies_inside_them:", int((cv > 1e-9).sum()),)
print("weights:_min_%.3f_ median_%.3f_ max_%.3f" % (cl.w.min(), cl.w.median(), cl.w.max()))
```

The weight is constant within every cluster, but not *across* clusters.

- Everyone in cluster selected with equal probability
- But weights differ *across* clusters, which tells us different cluster are selected with *different* probabilities.

1.25 Inferring sample design from weights

For household i in cluster c_i , a two-stage design factors the probability of selection:

$$\pi_i = \underbrace{p_{c_i}}_{\text{cluster chosen}} \times \underbrace{\frac{b_{c_i}}{M_{c_i}}}_{r_i: \text{ chosen within it}}$$

b_{c_i} is the number of households in the cluster selected (twelve); M_{c_i} is the size of the cluster.

1.26 What the weights reveal

The weight is proportional to the reciprocal selection probability:

$$w_i \propto \frac{1}{\pi_i} = \frac{M_{c_i}}{p_{c_i} b_{c_i}}$$

One equation, two unknowns. We see w and b ; we want p and M .

1.27 Inferring sample design from weights

```
print("sum_of_weights_over_the_whole_survey:", round(s["weight"].sum(), 1))
print("households_in_the_survey:", len(s))
```

Equal — the library normalizes weights to mean 1, so they carry *relative* probabilities and nothing about population totals. Everything below is a ratio.

Perhaps the fifty clusters were drawn by **simple random sampling** from Bafata's enumeration areas, each equally likely. Then $p_c = N/n$ is the same for all of them and the identity collapses:

$$w_i \propto \frac{M_i}{b_i}$$

A household in a large EA is less likely to be drawn, so it stands for more people.

```
cl["M_rel"] = cl["w"] * cl["take"]
cl["M_rel"] = cl["M_rel"] / cl["M_rel"].median()
```

```
print(cl["M_rel"].describe(percentiles=[.05, .25, .5, .75, .95]).round(2).t
```

Sanity check : Half the clusters fall between 0.89 and 1.04 times the median. Enumeration areas are usually *built* to be roughly equal in size.

1.28 Recovering both probabilities

```
cl["pi_rel"] = 1 / cl["w"] # overall, relative
cl["stage2"] = cl["take"] / cl["M_rel"] # within-cluster, relative
cl["stage1"] = cl["pi_rel"] / cl["stage2"] # what is left over
```

```
print("stage-1_probability_across_the_50_clusters:")
print("cv=%0.2e (SRS_says_it_should_be_constant)" % (cl["stage1"].std())
```

Zero to numerical precision. This is about the simplest case of two-stage cluster sampling:

1. EAs drawn with equal probability ($r_c = 1/\text{Number of EAs}$)
2. Households within a EA drawn with equal probability ($p_i = 1/M_{c_i}$)

1.29 Hazards of ignoring sample weights

Consider the following **Population Pyramid**, a nice graphical device for thinking about the demographic structure of a population. Because we're after population statistics, here we use the sample weights:

```
from lsms_library.visualizations import population_pyramid
```

```
ax = population_pyramid(glss, wave=WAVE, weights=True)
ax.figure.set_size_inches(7, 5.5)
```

1.30 Hazards of ignoring sample weights

Now, contrast if we *neglect* the survey weights, and just treat our data as if it was a simple random sample:

```
ax = population_pyramid(glss, wave=WAVE, weights=True, ghost=False)
ax.figure.set_size_inches(7, 5.5)
```

The filled bars are weighted; the outline is unweighted. The outline stands proud of the fill at every young age band and hugs it at the old ones, so the unweighted sample contains proportionally **more children** than the population it represents.

1.31 Why?

Let's consider two facts.

```
r = glss.household_roster(waves=[WAVE]).reset_index()
s = glss.sample(waves=[WAVE]).reset_index()
```

```
people = len(r)
weighted_total = s.set_index("i")["weight"].reindex(r["i"]).sum()
print(f"people_in_the_roster: {people}")
print(f"sum_of_their_weights: {weighted_total:.1f}")
print(f"mean_weight_per_household: {s['weight'].mean():.4f}")
```

The mean weight is 1 by construction. But that mean is over **households** while the roster is one row per **person**. So the weighted person-total is $\sum_h n_h w_h$, which equals the headcount only if weight and household size are uncorrelated.

So are they uncorrelated?

```
size = r.groupby("i").size().rename("hhsz")
d = s.set_index("i").join(size).dropna(subset=["hhsz"])
print("corr(weight, household_size)=", round(d["weight"].corr(d["hhsz"])))
print()
print(d.groupby(pd.cut(d.hhsz, [0, 1, 2, 3, 5, 8, 50],
                        labels=["1", "2", "3", "4-5", "6-8", "9+"]),
```

```

        observed=True)
    .agg(households=("weight", "size"), mean_weight=("weight", "mean"))
    .round(3).to_string())

```

Bigger households carry smaller weights. Since bigger households hold more children, an unweighted count over-represents the young.

2 Describing Welfare

2.1 Introduction

2.1.1 Reading

Deaton is in your `reading/` folder on the hub, or PDF.

- Deaton, chs. 3 and 4
- Deaton & Zaidi, *Guidelines for Constructing Consumption Aggregates*
 - Deaton & Zaidi §§2–4 if you read nothing else
- **Christiaensen, Ligon, & Sohnesen (2022)**: at least memorize the title.

2.1.2 Measuring Individual Welfare

Many notions of individual welfare; generally lead to some unidimensional index, say w (bigger is better):

- Subjective ("How happy are you?")
- Utility (How is this different from Subjective happiness?)
- Material (Something about command over economic resources)
 - Wealth?
 - Income?
 - Consumption expenditures?

1. Aside: What about risk?

2.1.3 Material Well-Being: Statistics

Some summary statistic of the *distribution* F of w in the population.

Inequality E.g., Atkinson's (1971) utilitarian notion of inequality:

$$I = U \left(\int_{\underline{w}}^{\bar{w}} w dF(w) \right) - \int_{\underline{w}}^{\bar{w}} U(w) dF(w) \geq 0$$

Poverty A quantile of some function $\phi_z(w)$; e.g.,

$$P(z) = \int_{\underline{w}}^z \phi_z(w) dF(w)$$

These are distributions *of the population*, so in practice those sample weights will matter!

2.1.4 Conventional options for w

1. Income

- Difficult to measure if sources are agricultural or informal
- Also highly variable—a farmer's main income may arrive once a year, after harvest, and be very risky.

2. Wealth

- What is it?
- Perhaps something like discounted expected income?
- But if income is difficult to measure then all possible future incomes may be much more difficult

2.1.5 Conventional options for w , continued

1. Assets

- Related to wealth, but may be measurable now
- Financial assets + Bicycles + Goats + House = ?? In principle these could be priced, but this may not be at all practical.

2. Consumption Expenditures The "Gold standard"

- Probably easier to measure than income
- Consumption is smoother than income
- And closer to the theoretical object (permanent income, lifetime resources) that we really want.

2.2 The Theoretical Object

2.2.1 The household's problem

What is the consumption aggregate a measure *of*? Assume:

- (A1) **Preferences** $U(c)$ over J goods, increasing, strictly concave, the same every period, discounted at β .
- (A2) **One decision maker** the household maximizes a single U .
- (A3) **One asset** a risk-free bond, gross return R , no borrowing limit.
- (A4) **Expectations** $\mathbb{E}_t$ over the true conditional distribution.

With A_t resources on hand and $x_t = p_t \cdot c_t$ total expenditure,

$$V_t(A_t, p_t) = \max_{c_t} U(c_t) + \beta \mathbb{E}_t V_{t+1}(A_{t+1}, p_{t+1}),$$

subject to $A_{t+1} = R(A_t - x_t) + y_{t+1}$.

2.2.2 The marginal utility of expenditure

Differentiate. With $u_j = \partial U / \partial c_j$, the first-order condition in c_j and the envelope condition in A are

$$u_j(c_t) = \lambda_t p_{tj}, \quad \lambda_t \equiv \beta R \mathbb{E}_t [\partial V_{t+1} / \partial A] = \partial V_t / \partial A.$$

So λ_t is two things at once, which is what makes it useful:

- the multiplier on the period budget constraint, the utility value of one more cedi spent *today*;
- the derivative of the *value function*, the utility value of one more cedi of resources, the whole future included.

V is concave in A , so λ falls as the household gets richer. Write $w_t = -\log \lambda_t$, which therefore rises with welfare. This is another candidate for the w of the opening slide.

2.2.3 The martingale condition

Substituting the envelope condition into the definition of λ_t ,

$$\lambda_t = \beta R \mathbb{E}_t \lambda_{t+1}.$$

When $\beta R = 1$ the marginal utility of expenditure is a martingale:

$$\mathbb{E}_t \lambda_{t+1} = \lambda_t.$$

A statement about information: whatever the household knew at t is already in λ_t , so nothing dated t forecasts the change.

- Needs A1–A4 *and* an interior solution: no binding borrowing limit.
- With several assets, $\beta \mathbb{E}_t[(\lambda_{t+1}/\lambda_t)R_{t+1}^j] = 1$ for every asset j held in positive quantity.

2.2.4 The martingale condition

Three things this doesn't say, since all three get attributed to it.

It isn't a statement about consumption. It's a statement about marginal utility, and getting from the one to the other takes a further assumption, which is the business of the next two slides.

It isn't a statement about $\log \lambda$. Jensen's inequality runs the wrong way: if λ is a martingale then $\mathbb{E}_t \log \lambda_{t+1} < \log \lambda_t$, so the welfare index $w = -\log \lambda$ drifts upward even when the model holds exactly. Anyone who regresses Δw on lagged information and recovers a constant has found the constant, not a violation.

And it isn't a statement about levels. $\beta R = 1$ is a knife edge adopted for convenience. Impatient households have $\beta R < 1$, so $\mathbb{E}_t \lambda_{t+1} = \lambda_t/(\beta R)$ exceeds λ_t : marginal utility drifts up, and consumption falls across the life-cycle.

Where the condition fails is more interesting than where it holds. Suppose the household would like to borrow and can't. Then A3 is false, the first-order condition holds as an inequality, and $\lambda_t > \beta R \mathbb{E}_t \lambda_{t+1}$: current marginal utility is too high relative to future, which is what being constrained means. Distinguishing that household from one that is simply saving against future risk is hard, since both look like a household that isn't spending. One approach elicits the intertemporal price $q = \mathbb{E}_t[\beta \lambda_{t+1}/\lambda_t]$ directly and asks how it moves in response to a windfall: up if a constraint was binding, down if precaution dominates. Not something an LSMS will tell you.

2.2.5 Why total expenditure?

Let $v(x, p) = \max\{U(c) : p \cdot c \leq x\}$, so $\lambda = \partial v / \partial x$.

At *given* prices x orders welfare, since $\lambda > 0$. Nothing beyond A1 is needed, but this ranks within a market only.

2.2.6 Ranking welfare across markets

To rank across markets we need a scalar $P(p)$ with $v(x, p) = \psi(x/P(p))$. Such a P exists if and only if U is homothetic (A5), and then $P = e(p)$, the cost of a unit of the aggregate. Quasi-homothetic (A5'): two indices, a line $a(p)$ and a deflator $b(p)$.

So $\lambda = \psi'(x/e(p))/e(p)$, decreasing in real expenditure.

Under A5, real total expenditure $x/e(p)$ is a sufficient statistic for within-period welfare. It is what the rest of this session builds.

2.2.7 Why total expenditure?

A5 is doing more work than it appears to. It is exactly the license to divide nominal expenditure by a price index, which is what the deflation slide does and what every poverty statistic in this session does after it. Better to carry it as a stated assumption than as a technicality.

It can be weakened, and the weaker version is the one this session's practice already presumes. Suppose preferences are *quasi*-homothetic (A5'): the expenditure function is Gorman's $e(u, p) = a(p) + b(p)u$, so $v(x, p) = \psi((x - a(p))/b(p))$ and demands are linear in x . Two price indices now do the work of one, and both have names this session already uses. $a(p)$ is the cost of a fixed bundle at local prices, which is what a poverty line is; $b(p)$ deflates whatever is spent above it. A5 is the case $a = 0$. Under the iso-elastic cardinalization of the next paragraph,

$$w = \frac{1}{\sigma} \log \frac{x - a(p)}{b(p)} + \log b(p),$$

log expenditure above the line. So an absolute line plus a deflator, the procedure the rest of this session teaches, is exact for a quasi-homothetic system rather than an approximation to a homothetic one; and unlike A5 it is consistent with Engel's law, since Gorman Engel curves are linear in x without being proportional to it. What it is not consistent with is a demand system of rank three, which is where session 4 goes.

Getting from an ordering to a number takes one more choice. Ordinal preferences fix v only up to an increasing transform, and λ is not invariant

to that transform, so a cardinalization has to be picked. Take the iso-elastic one, (A6): $\psi(G) = G^{1-1/\sigma}/(1-1/\sigma)$. Then $\lambda = (x/e(p))^{-1/\sigma}/e(p)$, so that

$$w = -\log \lambda = \frac{1}{\sigma} \log \frac{x}{e(p)} + \log e(p).$$

Within a market $e(p)$ is common to everyone, so w is an increasing affine function of log real expenditure, and two households in the same cluster are ranked identically by the marginal utility of expenditure and by log consumption. Across markets the price term enters additively. That equivalence is worth holding onto: the elaborate object and the simple one agree, *given* A5 and A6.

Durables take a separate assumption, (A7). A refrigerator is an asset, not a meal, so buying one is a portfolio transaction, and the single bond of A3 hides it. What belongs in U is the flow of services, which is why the aggregate imputes a use value rather than recording the purchase. Handling the rest of the bundle without tracking the durable stock alongside it requires the services of durables to be weakly separable from nondurable consumption. Assumed, not tested, and it is the reason the aggregate is described as covering nondurables and services.

2.2.8 Hall's random walk

The strict permanent income hypothesis adds:

(A8) Constant prices or a martingale in p independent of λ , which puts the walk in quantities.

(A9) Quadratic utility $U(x) = -\frac{1}{2}(\bar{x} - x)^2$, so $\lambda_t = \bar{x} - x_t$ is *linear* in expenditure.

(A10) $\beta R = 1$ with R constant and known.

The martingale in λ is then a martingale in x :

$$\mathbb{E}_t[\bar{x} - x_{t+1}] = \bar{x} - x_t \implies \mathbb{E}_t x_{t+1} = x_t.$$

Consumption is a random walk (Hall 1978): nothing dated t forecasts Δx_{t+1} , lagged income included. Testable, but it needs longitudinal data.

2.2.9 Hall's random walk

A9 should bother you, though not for the reason usually given. Quadratic utility isn't homothetic, but it is *quasi*-homothetic: demands are linear in x , the expenditure function is Gorman's $a(p) + b(p)\tilde{u}$, and that is the weakened A5 of the previous aside, so the two sit together perfectly well. What should bother you is which member of the class it is. Its $a(p) = p \cdot \bar{c}$ is the cost of the *bliss* bundle, with expenditure below it, where the Stone-Geary members put $a(p)$ at the cost of a subsistence bundle and expenditure above it; it has increasing absolute risk aversion, so richer households take less risk in absolute terms, which is backwards; and marginal utility reaches zero at $\bar{c}$. It is adopted for exactly one property, that marginal utility is linear in x , which is what turns a martingale in λ into a martingale in x . Certainty equivalence is a feature of the quadratic, not of the world.

The one-good form on the slide is for legibility, not necessity. Take a vector c and any positive definite A , and $U(c) = -\frac{1}{2}(\bar{c} - c)'A(\bar{c} - c)$. The first-order condition $A(\bar{c} - c) = \lambda p$ and the budget give

$$\lambda = \frac{a(p) - x}{b(p)^2}, \quad a(p) = p \cdot \bar{c}, \quad b(p)^2 = p' A^{-1} p,$$

which is linear in x at *any* prices, with the same a and b as the Gorman form above. So A8 is not a claim about how many goods there are. Nor, strictly, is it a claim that prices are constant: that is sufficient, since it makes a and b constants, but what the argument actually uses is that $\lambda_t p_t$ be a martingale, and prices that are themselves a martingale, with innovations independent of those in λ , deliver that too. The demands are $c_t = \bar{c} - \lambda_t A^{-1} p_t$, linear in λp , so the *quantity* vector is then a martingale, $\mathbb{E}_t c_{t+1} = c_t$, and so is consumption valued at any fixed base prices; which is Hall's object, since he had one good. What is not a martingale, even then, is nominal expenditure. At given λ , $x = p \cdot \bar{c} - \lambda p' A^{-1} p$ is concave in p , so a mean-preserving spread in prices lowers expected spending: $\mathbb{E}_t x_{t+1} = x_t - \lambda_t \text{tr}(A^{-1} \Sigma_p)$. Second order in price volatility, but a drift, and downward. A panel test of the random walk should therefore be run on consumption at constant prices rather than on nominal expenditure, which is the usual practice anyway and now has a reason.

Swap A9 for the iso-elastic A6 and the conclusion changes. The martingale is in $\lambda = (x/e)^{-1/\sigma}$, so $\mathbb{E}_t x_{t+1}^{-1/\sigma} = x_t^{-1/\sigma}$, and Jensen now delivers $\mathbb{E}_t x_{t+1} > x_t$ whenever next period's expenditure is risky. Consumption drifts up, because the household accumulates a buffer and runs it down slowly. That is precautionary saving, and it lives in the third derivative of U , which

the quadratic sets to zero. The random walk didn't survive a change of cardinalization, which is a reason to ask what else doesn't.

The empirical record is worse than that. Consumption growth is predictable from lagged income in most panels that have looked, the "excess sensitivity" of Zeldes (1989) and a large literature since, and the buffer-stock models of Deaton (1991) and Carroll (1997) fit low-income data considerably better than the strict version does.

What survives is the weak claim, and it is enough. The marginal utility of expenditures/assets λ_t is forward looking because it is the derivative of a value function, and it stays forward looking whether or not $\beta R = 1$ and whether or not utility is quadratic. A number built from this period's consumption therefore carries the household's own view of its prospects in a way that a number built from this period's income does not. That is the entire case for what follows. It is a good deal weaker than "consumption is a random walk," and fortunately it is all we need.

2.2.10 Assumptions about household choice

	Assumption	What it buys
A1–A2	Time-separable U , one maximizer	a well-defined λ
A3	One asset, no borrowing limit	an interior solution
A4	Rational expectations	$\mathbb{E}_t$ is the truth

2.2.11 Assumptions connecting choice to measurement

	Assumption	What it buys
A5	Homotheticity, or quasi-	a deflator, or a line and one
A6	Iso-elastic cardinalization	w affine in log real x
A7	Durables weakly separable	nondurables and services
A8–A10	Constant p , quadratic U , $\beta R = 1$	Hall's random walk

Everything from here to the end of the session is an attempt to measure x . None of it repairs these assumptions.

3 Elicitation of Consumption and Survey Design

3.1 Practical Difficulties in Measuring Individual Items of Consumption

3.1.1 The five components of consumption expenditures

Or: "Nondurable consumption and services".

Consumption expenditures in the LSMS surveys generally fall into one of five categories, and are typically elicited in different modules.

1. **Food** — purchased, own-produced, and received as gifts
2. **Non-food, non-durable** — fuel, transport, clothing, school fees, health
3. **Durables** — *value of services*, or "use value", not purchase price
4. **Housing** — rent, actual or imputed
5. **Other Services** — Not really measured at all

3.1.2 List Length

Beegle et al (2012): the *number* of goods asked has a large effect on recall, and on that evidence many NSOs have lengthened the list. Just for food:

Survey	Earlier rounds	Later rounds
Ethiopia	25 → 26 → 55	73 → 74
GhanaLSS	58 → 58 → 117 → 112	114 → 114 → 179 → 400
Burkina Faso	55 → 130	152
Uganda	61 → 63 → 70 → 71	77 → 77 → 117 → 116
Nigeria	90 → 90 → 96 → 104	116 → 116 → 110 → 110

- Should make measured total consumption more accurate;
- BUT: Really complicates comparisons across time.
- And: Can greatly increase cost of fielding the survey.

3.1.3 Timing & Recall

The model treats time as discrete. What in the field corresponds to a "period"? LSMS practice reveals what the designers thought.

Food Usually a seven-day recall, with interesting variation and some explicit experiments (Beegle et al. 2012) on how well respondents remember purchases.

Non-food expenditures or "semi-durables" Think things like clothing, inventories of kerosene. . . . Often a 30-day recall.

Durables Bicycles, radios, fans, Often a year-long recall for purchases, and/or an inventory of the *stock* of durables.

3.1.4 Asking about food: GhanaLSS 2016-17

At each of six visits, five days apart. *Maize, rice, cassava, garden eggs, smoked fish, cooking oil, . . .* — 179 items:

1. Food purchases

"How much was spent and what quantity was acquired of . . . since my **last visit**?"

amount, quantity, unit, per visit; a diary kept between visits.

2. Autoconsumption

"How much of own produced . . . was consumed by the household since my **last visit**?"

3.1.5 Asking about food: Uganda 2019-20

The case of Uganda is more standard than Ghana; there's a standard food expenditure module which is basically the same across the eight LSMS-ISA countries. For Uganda there are 129 food items in 2019-20.

3.1.6 Asking about food: Uganda's single visit

1. Single Visit

"How many [UNITS] of [ITEM] did your household consume out of [SOURCE] at home **during the last 7 days**?"

Then [SOURCE] takes values:

- "purchases"
- "purchases away from home"
- "out of home production"
- "received in-kind"

3.1.7 Asking about non-food: the window stretches

1. Ghana 2016-17 Asked once, at the last visit:

How much was spent on. . . in the past 12 months altogether?"

Frequently-bought non-food (toothpaste, haircuts, transport) also inside the six-visit food module instead.

3.1.8 Asking about non-food: Uganda 2019-20

1. Uganda 2019-20 Asked once:

"How much did the household spend in total on these [UNITS] of [ITEM] during the last [PERIOD]?"

PERIOD:

- 30 days for non-durables (e.g., soap, electricity, tires, stamps);
- 365 days for semi-durables (e.g., clothing, furniture, spoons, school fees).

3.1.9 Asking about durables: the stock, not the purchase

Attempts to assess the value of *stocks*, rather than flows.

1. Ghana 2016-17

"Does any member of the household own [ASSET]? How long ago was it obtained? What was its purchase price?
How much could you sell it now in Ghana cedis?"

2. Uganda 2019-20

"Does any member of your household own [ASSET] at present? How many? What is the **total estimated value** of the [ASSET] owned at present?"

3.1.10 Asking about housing: rent, actual and imagined

1. Ghana 2016-17

"How much does the household pay **in cash** towards the rent?"

An amount plus a time unit the *respondent* picks (daily ... yearly).
Owner-occupiers skip the block; no imputed rent is ever asked.

3.1.11 Asking about housing: Uganda 2019-20

1. Uganda 2019-20 Rent is part of the same 30-day recall on non-food expenditures:

"How much did the household spend in total on these [RENT OF RENTED HOUSE] during the last 30 days?"

Part of the same module as non-durables; same question format.

"What would be the total value of [IMPUTED RENT OF OWNED HOUSE]... if you were to sell at the farm gate, during the last 30 days?"

3.1.12 Food prices: which price?

Food *expenditures* are quantities times prices. Which prices?

- farmgate price received — what the household gave up
- local market price — what it would have cost
- these differ by the marketing margin, said to be 30–50%

Deaton & Zaidi: value at the price the household *faces*. In practice, a median unit value by item, cluster and round, falling back up the hierarchy when cells are thin.

Which of the two does a survey's own valuation question measure?

3.1.13 Food prices: what GLSS7 asks, and what it gets

Section 8 Part H, question 9: "**for how much would you sell one unit of ... now?**" A farmgate question, by its wording (though hypothetical).

Against what neighbours *paid*, same item, unit and cluster:

- Ghana 2016-17, 322 cells: median ratio **1.000**; a third tie exactly
- Uganda, eight waves, the same question: **1.000** pooled
- Uganda also records sales: the valuation is **1.23** times the sale price; buyers paid **1.60** times it

Asked what they would sell a bowl of maize for, a household answers with the price of a bowl of maize. The margin is real — 1.3 to 2.2 where a survey records sales.

3.1.14 Food prices: which price?

Start with what is actually there. Four sources get named in discussions like this one — unit price, market price, farmgate price, community price — and GhanaLSS 2016-17 has three, of which two of the names refer to the same instrument. `food_prices()` is the unit value implied by an expenditure, and since `lsms_library` 0.14.0 that is no longer purchases alone: the library derives an expenditure for own production, and dividing it by the quantity returns the household's own reported price wearing a different hat. The `s` level is what tells the two apart, and it is not a formality — at the default `units='kgvalue'`, **three Ghanaian waves (1998-99, 2005-06, 2012-13) come back with produced rows and no purchased rows at all.** Subset on `=s` before comparing anything here with anything, or you will compare the valuation with itself. `community_prices()` is the community questionnaire, priced at the cluster, which is both the "market" and the "community" price of the list. And own-production rows in `food_acquired()` carry a price the household itself gave, elicited as question 9 of Section 8 Part H: "for how much would you sell one unit of ... now?" By its wording that is a farmgate question, what the household would *receive*, so the comparison the slide sets up ought to be available.

Measured, the elicitation for the value of consumption out of own production is not in fact a farmgate price. Put the household's valuation beside the unit value its neighbours *paid* for the same item, in the same unit, in the same cluster, keeping only cells with at least ten observations on each side, and the median ratio across 322 cells is 1.000; 31.7% of cells agree to the pesewa. Neither is this a Ghanaian peculiarity. Uganda asks the same question in eight waves, and pooling the two countries gives 1,666 cells with a median ratio of 1.000. *Asked what a bowl of maize would sell for, a household answers with the price of a bowl of maize.*

Uganda is where to see what that costs, because it is the one survey that also records actual crop sales. On the 223 cells where all three prices exist, buyers paid 1.60 times what sales fetched — there is the marketing margin — and the household’s valuation was 1.23 times the sale price, in between and no nearer one end than the other. The interviewer manual there says in so many words that the valuation is to be at the farm-gate price, excluding transport and marketing. It is not. So an aggregate that values own consumption at the price in `food_acquired` is valuing it about 23% above what the household would actually have received, and carries part of a margin the questionnaire meant to exclude. Ghana records no sales, so the same number cannot be produced here; but the valuation behaves the same way, and there is no reason to expect the consequence to differ.

The margin itself is real. Where a survey records both a sale and a purchase of the same good in the same container, the ratio runs from 1.30 in Ethiopia through 1.50 in Malawi and 1.52 in Nigeria to 1.57 in Uganda and 2.24 in Tanzania: the textbook’s 30–50% and then some. What the textbook does not say is that the question surveys use to elicit a producer price does not measure one.

The community questionnaire tells a weaker version of the same story. Its price sits 20% above the household’s valuation in 2016-17, 5% and 109% above it in 1991-92 and 1998-99, and 13% below it in 2012-13, on a few dozen cells each, so there is no stable margin there either; the enumerator weighs what a vendor sells and the household reports a container, and the two units seldom mean the same thing. Whatever the household reports, it tracks what its neighbours pay far more closely than what a vendor charges.

What follows for practice. Deaton and Zaidi treat the choice between the producer price and the market price as the analyst’s. It is not: the respondents have made it, and they chose the market. Value own production at `food_acquired.Price` if you like, since at the median it is the price the household faces and that is what the guidelines ask for; but do not call it a farmgate price, and do not expect it to have moved your aggregate down by a third. If you need what production would fetch, you need a survey that records sales, and GLSS7 does not do that.

The numbers above are from a study of eight countries’ price sources in `LSMS_Library`, with the cells built to a common protocol; `price_sources.ipynb` builds Ghana’s three prices at a simpler grain and asks what the data can and cannot say about them.

3.1.15 Food quantities, and what was never bought

Much of the food in an LSMS household was never bought:

- Purchased at market
- Consumed out of own production
- Gift/Transfer
- Consumed outside the household (?)

Not a canard. In Ghana 2016-17 own production is **28.7%** of the value of food; 12,191 of 13,942 households record some. Deaton and Zaidi's Table 3.1 implies 31.9%.

But it reaches your aggregate only if you *value* it: from 1991-92 on, the survey records own production in quantities, not cedis.

3.1.16 Food quantities, and what was never bought

The claim survives contact with the data, which is not something one can assume of a claim repeated as often as this one. Valuing own production at the price the household itself reported, the share of food value that was never bought runs across the seven Ghanaian waves as

wave	1987-88	1988-89	1991-92	1998-99	2005-06	2012-13	2016-17
%	38.8	33.4	26.7	30.5	17.1	27.6	28.7

The two 1980s rounds are the highest; from 1991-92 the share settles between a quarter and a third and stays there, so whatever is happening it is not a steady decline over thirty years. The 2005-06 figure is the one to be suspicious of, and chasing it down would make a decent exercise on its own.

Until `lsms_library` 0.14.0 that table was a calculation you had to perform yourself, because the library served own production without an expenditure from 1991-92 on and `basis'total'` = silently returned purchases alone. It now derives the expenditure and stamps the rows it computed, so `basis'total'` = carries own production in six of the seven waves. 1991-92 is deliberately left alone: its dominant unit is A11, "the whole harvest", which is not a quantity a price can multiply.

Two things are worth knowing before you trust the 28.7%. The first is that the price is the household's own, not an imputation of ours, and it varies household by household within an item and unit: seven distinct values in the

median item-unit cell, seventy-seven in the worst. So it is real reported data. Comparing it against the unit values implied by *purchases* of the same item in the same unit, the median ratio across 178 cells is 1.00, which is what the previous aside leads you to expect: the household is quoting the market price. The tails are not reassuring at all, running from 0.5 at the tenth percentile to 4.0 at the ninetieth, so while the aggregate may be sound any individual household's imputation may not be.

The second is that 28.7% is a floor. In 2016-17 the survey records a price on 151,683 own-production rows but a quantity on only 138,793 of them, and a row missing either one contributes nothing to the product. Note also that GhanaLSS distinguishes two sources, purchased and produced, so the gifts and the meals eaten away from home in the list above are nowhere in this particular table. Both omissions push the same way.

The form that produces these numbers is in your `reading/` folder as `GLSS7-2016-17-Section8H-own-produce.pdf`: quantity consumed at each of the six visits, and one price, asked as what a unit would sell for now. The exercise notebook `food_sources.ipynb` takes this apart, and asks whether the pattern holds in the countries you care about.

3.1.17 Building the aggregate

LSMS_Library does the food side for you.

1. Preface

```
# Show tracebacks without the library's internal frames: the line that
# failed, and why. Change Plain to Verbose if you ever want the rest.
%xmode Plain

# The library audits its own corpus on first read and reports what it j
# — implausible quantities, NaN index keys, a column that is wholly i
# one wave — at multi-paragraph length. Those reports are a work qu
# whoever maintains the data, not something the room can act on, and th
# the output they are attached to. Silenced here by their own
# "Set LSMS_..._STRICT=1" signature, which is precise: every other warn
# pandas deprecations included, still shows. Delete these two lines to
# them.
import warnings
warnings.filterwarnings("ignore",
                        message=r"(?s).*Set_LSMS_[A-Z_]+=1_to_make_this
```

*# The first call that builds a table may print "DVC unavailable ...
 # falling back to manual aggregation". That is about how the library
 # fetches its raw files, not about your data; the numbers are the same,
 # and it does not recur once the table is built.*

```
import matplotlib as mpl
mpl.rcParams.update({'figure.dpi': 110, 'font.size': 11,
                    'lines.linewidth': 1.8})
```

```
import lsms_library as ll
import numpy as np, pandas as pd
```

2. Construct data on food expenditures What features are there?

```
ll.features()
```

What countries have `food_acquired`?

```
ll.countries(feature='food_acquired')
```

Choosing a country from the list of those that have the `food_acquired` feature:

```
USE_COUNTRY = 'GhanaLSS'
```

```
centry = ll.Country(USE_COUNTRY)
```

```
centry.food_acquired()
```

The index is $(t, i, j, u, s, visit)$ — wave, household, item, unit, source, visit — and the columns are **Expenditure**, **Quantity**, **Price**. The derived table collapses this to expenditure per household-item:

```
food = centry.food_expenditures()
food.head()
```

Summing over items gives food *purchases* per household. Not consumption:

```
hh_fx = food.groupby(['t', 'i', 'v']).sum().squeeze()
hh_fx.groupby('t').describe().round(0)
```

Note that these purchases always default to being in units of **LCUs** (local currency units). We can see the units if we ask:

```
hh_fx = cntry.food_expenditures(currency='index').groupby(['t', 'i', 'v'])
hh_fx.groupby(['t', 'currency']).describe().round(0)
```

```
hh_fx = cntry.food_expenditures(numeraire='LCU-real-2017')
```

3. Construct data on food out of own production (autoconsumption). These expenditures don't include autoconsumption (consumption out of own production). For the GLSS7 those are elicited in a different section: asks amount and quantity and never a price.

We can include value of all food acquired (included out of production):

35

No value of own production here? Look at **Expenditure** on the **produced** rows before you assume it: the library fills it in every wave but 1991-92, at the household's own reported price, and the **Derivation** column names the rule that filled it. Where it is still missing, do the multiplication yourself. `min_count=1` keeps a wave that has nothing to sum reported as missing rather than as a zero.

```
value = fa[ 'Expenditure' ].fillna( fa[ 'Quantity' ] * fa[ 'Price' ] )

by_source = value.groupby( [ 't' , 's' ] ).sum(min_count=1).unstack( 's' )

by_source[ 'Auto_share' ] = by_source[ 'produced' ] / by_source.sum( axis=1 )

by_source
```

Own production is 28.7% of the value of food in 2016-17. `value` is expenditure wherever one was recorded or derived, and quantity times the household's own price where neither was; summed over items it is food consumption per household, purchased and own-produced, and it is the aggregate the rest of this session uses.

3.1.18 GLSS7's answer: a diary, visited six times

The most recent GhanaLSS is ambitious about frequent purchases: a diary, and an enumerator visiting six times at five-day intervals.

The form (**reading/**): six visit columns headed *2nd* to *7th*, asking **"...since my last visit?"**, and no price column anywhere.

Expensive; two experiments have asked whether it is worth it.

Scott & Amenuvegbe (1990) Ghana, and the reason GLSS asks this way

Beegle et al. (2012) Tanzania, eight modules, randomly assigned

3.1.19 The diary

CLUSTER NUMBER:

HOUSEHOLD NUMBER:

NAME OF HOUSEHOLD HEAD: DATE:

A R DAY MTH Y E

INSTRUCTION: RECORD ALL PAYMENTS MADE AND ESTIMATE ALL OWN PRODUCED GOODS CONSUMED.

Date Amount	Place of Purchase/ Own Produce	Description of Item Purchased or Consumed	Qty	Unit	Price

"Record all payments made and *estimate all own produced goods consumed*": the diary asks for a price on own produce. Section 9B, filled from it, does not.

3.1.20 GLSS7's answer: a diary, visited six times

Worth putting the form beside the data, because it settles two things the code can only suggest.

The first is why purchased food has no price. The column headings are AMT, QTY and UNIT, repeated for each of the six visits, and there is nowhere to write a price. So the unit value you compute later is an arithmetic consequence of the form, not a choice anyone made, and the asymmetry with own production, which is asked at a price and never at an expenditure, is built into two different instruments rather than into the processing.

The second is the wording, "since my last visit". Every visit asks it, so the recall window is whatever the enumerator's schedule made it, and the household is not told what period to think about. Five days is the design; the form does not say five days. That is also the most likely explanation for the first visit sitting 20% above the rest, since the gap before the first expenditure visit is the one that is neither five days nor known. The exercise notebook asks you to test that against a longer-window story, and the form is why the question is worth asking rather than being a fishing expedition.

It is the same phrase, incidentally, that the 1987-88 and 1988-89 rounds record as their entire recall period. The wording did not change; what changed is that later rounds pinned it down with a visit schedule.

3.1.21 GLSS7's six visits

This is the form the diary is kept on, Section 9B: six visit columns headed *2nd* to *7th*, AMT, QTY and UNIT under each, and nowhere to write a price. Open it beside the code: Section 9B (PDF, opens in a new tab; also **reading/** in your home directory), and the booklet the household keeps between visits, which 9B is filled from: the diary.

GLSS7 — Section 9 Part B: Expenses (frequently purchased items)

Ghana Living Standards Survey round 7 (2016/17), household questionnaire Module B: the opening of the food expenditure elicitation. Note the six visit columns (2nd-7th), the recall wording "since my last visit", and that the answer columns are AMT, QTY and UNIT: there is nowhere to record a price. Source: Ghana Statistical Service, GLSS7, MODULE B_7A_7D_26.docx, from www2.statsghana.gov.gh/catalog/37. Transcribed from the Word original for teaching: this is not a scale of the printed form.

PERSON RESPONSIBLE		ID																	
COPY FROM SECTION 6 D.8																			
2nd		3rd		4th		5th		6th		7th									
VISIT																			
day month day month day month day month day month day month day month																			
ITEM	C O D E	1.			2.			3.			4.			5.			6.		
		How much was spent and what quantity was acquired of.....since my last visit?			How much was spent and what quantity was acquired of.....since my last visit?			How much was spent and what quantity was acquired of.....since my last visit?			How much was spent and what quantity was acquired of.....since my last visit?			How much was spent and what quantity was acquired of.....since my last visit?			How much was spent and what quantity was acquired of.....since my last visit?		
		AMT (GHS)	QTY	UNIT	AMT (GHS)	QTY	UNIT	AMT (GHS)	QTY	UNIT	AMT (GHS)	QTY	UNIT	AMT (GHS)	QTY	UNIT	AMT (GHS)	QTY	UNIT
FOOD AND NON-ALCOHOLIC BEVERAGES																			
Rice - Local																			
	001																		
	002																		
	003																		
	004																		
	005																		
	006																		
Rice - Imported																			
	010																		
	011																		
	012																		
	013																		
	014																		
	015																		
	016																		
	017																		
	018																		
	019																		
	020																		
	021																		
	022																		
	023																		
	027																		
Guinea corn/sorghum																			
	028																		
	029																		
	030																		
	031																		

The diary design is visible in the data. GLSS7 carries a **visit** level:

```
glss7 = cntry.food_acquired().xs('2016-17', level='t').xs('purchased', level='i')
sorted(glss7.index.get_level_values('visit').unique())
```

Six of them, five days apart. The form numbers them 2 to 7, because the food questions begin at the second visit; the library on the hub labels them 1 to 6, and other versions use the form's numbers. The text below uses the form's numbering, and the code selects the first food visit by position so that it does not care. Each visit covers the same five days, so if the instrument worked perfectly they should all look alike. They do not:

```
byvisit = pd.DataFrame({
    'exp': glss7.groupby(['i', 'visit'])['Expenditure'].sum(),
    'items': glss7.groupby(['i', 'visit']).size(),
```

```

})
(byvisit.groupby('visit')
  .agg(mean_exp=('exp', 'mean'), mean_items=('items', 'mean'))
  .assign(exp_per_item=lambda d: d.mean_exp / d.mean_items)
  .round(2))

```

The first visit stands well above the five that follow, and then the series is flat. That is the shape Scott and Amenuvegbe found: steep early, flat later. Which of two stories is it, though? More items remembered, or bigger amounts attached to the same items?

```

v1 = sorted(byvisit.index.get_level_values('visit').unique())[0]
# first food visit
first, rest = byvisit.xs(v1, level='visit'), byvisit.drop(v1, level='visit')
for col in ('exp', 'items'):
    print(f"{col:6s} first food visit vs the other five: "
          f"{100*_((first[col].mean()-rest[col].mean()-1):+.1f}%")

```

Expenditure is 20.6% higher; the item count only 4.5%. So it is mostly larger amounts per item rather than fuller enumeration, which is what telescoping looks like: purchases from before the window get pulled into it. It is the start-up bias Scott and Amenuvegbe had to strip out before they could measure decay at all, and here it is in the survey you are about to build an aggregate from.

Whether the first visit is worth keeping is a real question, and `recall_and_diaries.ipynb` is where you get to argue about it.

3.1.22 How fast does recall decay?

Scott & Amenuvegbe (1990), 13 frequently purchased items in Ghana:

- Reported daily expenditure falls steadily as the recall window lengthens from one day to one week, then flattens between one and two weeks
- Roughly **2.9% lost for every day added**, after reweighting away purchase cycles and start-up bias
- The decline is larger for items bought more often
- A one-year recall is simply erratic

So a weekly recall understates frequent purchases by around **20%** — and that was, in their words, "an abnormally well controlled field situation."

3.1.23 How fast does recall decay?

The shape matters as much as the size. The decay is steep early and flat later, not linear, which is why the choice between a one-day and a one-week window costs far more than the choice between one week and two. A survey designer reading only the headline number would conclude that shorter is always better and keep shortening; the curve says the returns are nearly exhausted after a week, and that the remaining error is not recall at all.

It is worth dwelling on that last phrase, "an abnormally well controlled field situation." Scott and Amenuvegbe are describing their own experiment, not a production survey, and they raise the 20% as a lower bound on what a routine operation should expect. Anyone quoting the figure as *the* recall bias is quoting it a good deal more confidently than its authors did.

3.1.24 What actually drives the error

Beegle et al. (2012): eight modules randomly assigned across ~4,000 Tanzanian households, benchmarked against a personal diary.

- Recall *length* is the small knob: 7 vs 14 days moves food spending **7%**
- Item *list* design is the big one: **21%** between 58 items and 11 categories
- "Usual month" fails and contaminates the next module
- Household diaries miss about **19%** of what personal diaries catch

Poverty ranges from **47.5% to 66.8%**, by module alone.

3.1.25 What actually drives the error

The design deserves stating, because the comparisons only mean something against it. Five recall modules and three diary modules: a 14-day long list of 58 items; the same list at 7 days; a 7-day subset of 17 items covering about 77% of food expenditure and then scaled up; a 7-day list collapsed from 58 items into 11 categories; a "usual month" asked over 12 months; a household diary with frequent supervision; the same with infrequent supervision; and a personal diary, frequently supervised, as the benchmark.

Three findings are worth carrying away.

The first is that the profession has been tuning the wrong parameter. Recall length is what gets argued about, and it is worth about 7%; list design is what gets treated as an administrative detail, and it is worth three

times as much. The subset list is the practical discovery here: 17 items, rescaled, tracks the 58-item list closely while saving under ten minutes of interview time. Collapsing the list into categories saves about as little and costs 21%, so it is the one option that is simply dominated.

The second is that a bad module damages its neighbours. The "usual month" module reports lower food than a 14-day recall and nearly doubles interview time, and its *non-food* answers run 25% above one comparison module and 32% above another, even though the non-food questions were word-for-word identical across every recall module. The cognitive load of a hard food module spills forward into whatever follows it. Questionnaire order is not a neutral administrative choice.

The third is that the bias is not a scalar you can subtract off. Recall modules under-report most for poorer households and for households with more adults, while apparent telescoping rises with asset wealth, so the error is correlated with exactly the things being studied. Squared poverty gaps for the collapsed-list and usual-month modules come in at more than twice the benchmark. Against all that, household consumption *rankings* are "remarkably stable" across modules, which is the one genuinely reassuring result in the paper, and the reason relative statements survive a change of instrument when absolute ones do not.

Their operational advice, which is unusually direct for a paper of this kind: if you must pick one recall module, use a long list over 7 days; never collapse the item list, because the savings are trivial against the bias; treat the subset list as an acceptable substitute if you can afford to lose the detail; drop "usual month" entirely; use infrequent supervision for household diaries unless literacy is low, where it "will dramatically underestimate consumption"; and pay for personal diaries only where the population is heavily urban, since they cost six to ten times a recall module. They also caution that the whole thing is setting-dependent. Rural Tanzania generalises to much of rural Africa and South and Southeast Asia; urban middle-income settings, much less so.

3.1.26 Recall periods actually in use

What the surveys in `LSMS_Library` actually do, which is a good deal less uniform than the literature's advice would suggest:

country	wave(s)	recall
Ethiopia	2011-12 ... 2021-22 (5)	past one week
Togo	2018	7 days
Guatemala	2000	last 15 days
Panama	1997, 2003, 2008	last 15 days
Nigeria	2010Q3 ... 2016Q1 (6)	7-day consumption
Nigeria	2018Q3, 2019Q1	7-day produced, 30-day purchase
Tanzania	2008-09 ... 2020-21 (6)	consumed at home during...
Mali	2014-15	7 days
Serbia	2007	7-day diary
Burkina Faso	2014	7 days x 4 passages
Burkina Faso	2021-22	7 days and 30 days
GhanaLSS	1998-99, 2012-13, 2016-17	5 days x 6 asks
GhanaLSS	1991-92	2 days rural / 3 urban
GhanaLSS	1987-88, 1988-89	since my last visit

Note the two GhanaLSS entries at the bottom. "Since my last visit" is not a recall period at all, it is a recall period that varies by household and by enumerator schedule, and nothing in the delivered data records what it was. Comparing 1987-88 against 2016-17 therefore compares two instruments as much as two Ghanas, which is the point the list-length slide made earlier and the reason it is worth making twice.

3.1.27 Timing: what is a period?

For food, a day, probably: people schedule around meals. Surveys pick something longer, and each choice buys one error with another.

- A *diary* of what was eaten, meal by meal, is the ideal and the burden: hard to sustain, and paying for it buys a record, not an accurate one
- A *short* window is cheap but noisy: day-to-day variation in intake is not the error we are after
- *Usual* consumption asks the respondent to do our averaging for us, and to share our idea of "typical"; Beegle et al. found it fails outright

3.1.28 Inventories: purchases are not consumption

Storable staples — dry maize, rice, oil — are bought in bulk and drawn down. Perishables — fruit, fresh fish — are bought as eaten.

- A short window catches the fruit and misses the maize sack, or catches the sack and triples the week
- Averages out across households if purchase timing is unrelated to the visit; does not average out *within* one, which is where the line is applied
- Who buys in bulk is a matter of wealth: cash on hand, a dry store, a refrigerator. So the noise is on the rich, and on the staples

3.1.29 Measuring consumption despite inventories

Two approaches:

- Repeated visits over thirty days, as in GLSS7.
- Ask what was *consumed*, as Section 8H does, rather than only what was purchased, as in Section 9B.

Both address the mismatch between the timing of purchases and consumption.

3.1.30 Timing and inventories

The most natural "period" for food is probably a day; most people schedule around a certain number of meals per day. Some surveys implement a **food diary** method that tries to capture exactly what was eaten at different meals each day (either by individuals or by the household collectively). But the hassle of keeping up such a diary over an extended period is considerable, so it's difficult to induce respondents to keep an accurate record (paying them may tend to produce an *inaccurate* record).

What about a diary over a **short** period? Then the worry is that natural variation in food intake at such high frequencies will create a different form of measurement error, if what we're after is something like average intakes over time.

A handful of surveys try to have it both ways: they'll ask about what food was consumed in the most recent period (day/week/month), and then also ask about how much food is consumed in a *typical* period. This leans heavily on respondents *knowing* this, and also on having a shared understanding of what *typical* means.

Inventories are the part of this that the recall literature mostly skips. What a purchase module records is acquisition, and acquisition is consumption only for goods that cannot be kept. Fresh fish and fruit are bought as

eaten; dry maize, rice and cooking oil are bought in quantities meant to last, and drawn down. Over a week, then, the perishables are measured about right and the staples are measured as either nothing or a month's worth, depending on whether the sack happened to be bought inside the window. If purchase timing is unrelated to the visit this averages out across households, which is why the population mean survives; it does not average out within a household, and it is the within-household number that gets compared with a poverty line. And the lumpiness is not randomly assigned. Buying in bulk takes cash on hand, somewhere dry to keep a sack, and for anything that spoils a refrigerator, all of which come with wealth; so richer households' food expenditure is measured with more noise than poorer households', and the noise sits on exactly the staples whose budget share Engel's law makes the marker of poverty. Two designs address it. GLSS7's six visits over thirty days lengthen the window until most sacks fall inside it, at the cost the previous slides describe. A consumption module asks what was eaten from stock as well as what was bought, at the cost of having to value what was drawn down, which is the own-production problem under another name. The object the aggregate is after is consumption; Section 8H asks for it, for own produce, and Section 9B does not, for purchases.

3.1.31 Price Indices

Two different price indices, meant to address two different problems:

- **Temporal:** prices rise between rounds. Use the CPI, or better, a survey-based Paasche index built from the survey's own unit values.
- **Spatial:** prices differ across regions *within* a round, often by more than they differ across a decade. A national poverty line applied to undeflated nominal consumption will find poverty wherever prices are high.

3.1.32 A household-specific Paasche index

Deaton & Zaidi recommend a household-specific Paasche index

$$P_i = \left[\sum_j s_{ij} \left(\frac{p_{ij}}{p_j^0} \right)^{-1} \right]^{-1},$$

with s_{ij} household i 's budget share on item j .

3.1.33 Deflation

Spatial deflation is the step most often skipped and most often decisive. In a country the size of Ghana, the price of a fixed food basket can differ by 30% between Accra and the Upper East, which is comparable to the entire national increase in real consumption over a decade. An analysis that deflates over time but not over space will conclude that the north is poor for reasons that are partly an artefact of the accounting.

It also matters that the index is *household-specific*. A regional price index applies the same deflator to everyone in the region, and so implicitly assumes they consume the same bundle. They do not: the poor spend more of their budget on staples, so the relevant price index for the poor moves with staple prices. This is not a refinement; it is what makes the poverty rate respond correctly to a maize price shock.

3.1.34 Own production and the valuation problem

How much rides on this depends on where you are, and the range is wide. Deaton and Zaidi's Table 3.1 gives shares of the consumption aggregate:

Country	Food	of which purchased	of which home produced
Nepal	64.2	29.0	35.2
Ghana	65.2	44.4	20.8
Vietnam	50.9	34.1	16.8
Kyrgyz Rep.	44.5	33.4	11.1
Brazil	27.7	21.0	6.7
Panama	45.9	39.8	6.1
Ecuador	49.6	44.3	5.3
South Africa	30.4	28.2	2.2

So home production is a fifth of everything a Ghanaian household consumes and a third of its food, against a fiftieth in South Africa. Deaton puts the upper end higher still: in much of West Africa, he writes, "home production and hunting may account for a large share — perhaps more than a half — of total consumption" (ch. 1, p. 29). A quantity that large is not a correction to the aggregate; for these households it substantially *is* the aggregate, and it arrives entirely by imputation.

Which is worth stating the other way round: the more of the aggregate you impute, the less you are measuring and the more you are assuming. Deaton and Zaidi make the point that imputed values are less variable than

the truth would be, so leaning on them tends to *understate* inequality and poverty — the errors do not merely add noise, they push in a direction.

Deaton's example of imputation gone wrong is worth the retelling. A survey in West Africa recorded the water households fetched from local rivers and ponds, and priced every such item by the nearest traded equivalent. River water was therefore valued at the going price of Perrier in the nearest city, "thus endowing rural households with immense (but alas illusory) riches." Deaton and Zaidi return to it in a single dry clause: "water from the local pond is certainly different from L'Eau Perrier." The joke has a point — the algorithm was applied consistently, which is exactly what let it produce a consistent absurdity.

The unit-value hierarchy is worth dwelling on, because it is the first place in the workshop where a defensible answer requires a judgment rather than a formula. You have, for a given item, a set of reported (expenditure, quantity) pairs from households that bought it. Their ratio is a unit value, not a price: it embeds quality choice, package size, and reporting error, and a rich household's unit value for "rice" is higher partly because they are buying better rice.

The standard move is to take the median within the smallest cell with enough observations — cluster, then district, then region, then national — and use it for the households that did not buy. The median rather than the mean because unit values have a long right tail from misreported quantities. The hierarchy because the cluster cell will often be empty. And you should report how far up the hierarchy you had to climb, since an aggregate built mostly on national medians is a different object from one built on cluster medians.

3.1.35 Durables and housing

Durables: what enters consumption is the *flow of services*, not the purchase. For a stock S with service life L and interest rate r ,

$$\text{use value} \approx S \left(r + \frac{1}{L} \right).$$

Housing: for owner-occupiers there is no rent, so impute it.

- hedonic regression of log rent on housing characteristics, among renters
- predict for owners
- fragile where the rental market is thin, which is most rural areas

3.2 Intra-Household Allocation

3.2.1 Adult equivalence

A household of six is not six times as well off as a household of one with the same total consumption. Two adjustments, often conflated:

- **Household composition:** children cost less than adults $\Rightarrow$ adult-equivalent scale $A_i = \sum_k a_{\text{age,sex}}$
- **Economies of scale:** shared goods, so cost rises less than proportionately $\Rightarrow n_i^\theta$, with $\theta \in (0, 1)$

Combined: $c_i = \frac{C_i}{A_i^\theta}$.

θ is not estimated in this session; it is *assumed*. Report the poverty rate for several values of it.

3.2.2 Barten: children re-price goods

A^θ makes a child a fraction of an adult *for everything*. Barten (1964): composition changes *which* goods a household needs,

$$\max_c U\left(\frac{c_1}{m_1(z)}, \dots, \frac{c_J}{m_J(z)}\right) \quad \text{s.t.} \quad \sum_j p_j c_j = x,$$

so a household with demographics z is the reference household at prices $p_j m_j(z)$. A child makes milk dear; substitution does the rest.

3.2.3 Barten: what the scale depends on

- the scale is a ratio of cost functions, $e(u, p \circ m(z))/e(u, p)$: it depends on prices, and in general on u
- A^θ is the case where every m_j is the same number: children change the *size* of the bundle, never its *shape*
- a shape that depends on z is not consistent with homotheticity

3.2.4 Barten: children re-price goods

Barten's device is to put demographics into preferences good by good: effective consumption of good j is $c_j/m_j(z)$, so a household with children needs $m_j(z)$ units of j to get what the reference household gets from one. Since

$\sum_j p_j c_j = \sum_j (p_j m_j)(c_j/m_j)$, the household solves the reference household's problem at prices $p \circ m(z)$, and every demographic effect on the bundle is a price effect: a child makes milk dear relative to beer, and the household substitutes accordingly. The equivalence scale, the ratio of what it costs the two households to reach the same u , is $e(u, p \circ m(z))/e(u, p)$. In general that depends on prices and on u ; Lewbel's "independence of base" is the assumption that it does not depend on u , and it is what lets a scale be quoted as one number.

Now put this beside A5. Homotheticity says that at given prices every household consumes the same bundle up to scale, so a single price index $e(p)$ deflates everyone's expenditure. Barten says the shape of the bundle depends on z . The two are reconciled only by moving the homotheticity to *effective* consumption, U homothetic in $c/m(z)$. That buys independence of base, since the scale becomes $e(p \circ m(z))/e(p)$ with no u in it, but it costs the single deflator: the right index for a household of type z is $e(p \circ m(z))$, one per demographic type, and the scale is a ratio of price indices that moves when relative prices move. Cobb–Douglas is the one case where it does not, $e(p \circ m) = e(p) \prod_j m_j^{\beta_j}$; and in that case the m_j drop out of demand altogether, since $c_j = \beta_j x / p_j$ whatever z is, so the scale can be anything at all as far as the data are concerned. That is the general point in a special case. Barten scales are identified from how budget shares respond to the effective prices, so they are identified only to the extent that shares respond to prices at all; and under A5 shares do not respond to x , so the Engel and Rothbarth methods, which find the x at which two households have the same food share or the same adult-goods spending, have nothing to work with either. Pollak and Wales (1979) made the deeper point that demand data identify only *conditional* scales in any case, the cost of a child given that one has the child; a welfare comparison needs the unconditional one, and that has to come from an assumption, which is why θ is assumed rather than estimated on the previous slide.

3.3 Constructing Aggregate Welfare Measures

3.3.1 Poverty measures

The Foster–Greer–Thorbecke class: for poverty line z ,

$$P_\alpha = \frac{1}{N} \sum_{i:c_i < z} \left(\frac{z - c_i}{z} \right)^\alpha.$$

- $\alpha = 0$: headcount ratio. Ignores *how* poor the poor are.

- $\alpha = 1$: poverty gap. The transfer needed, per capita, as a share of z .
- $\alpha = 2$: squared gap. Weights the poorest most; the only member of the family that satisfies the transfer axiom.

Using sample weights: $P_\alpha = \sum_i w_i \left(\frac{z-c_i}{z}\right)^\alpha / \sum_i w_i$.

3.3.2 Poverty, correctly weighted

```
sample = cntry.sample()
```

```
def fgt(c, z, alpha=0, w=1):
    """
    Weighted FGT_alpha.
    c: consumption per adult equivalent;
    z: poverty line;
    w: weights.
    """
    # A scalar broadcasts and a Series reindexes, so 'w=1' (unweighted) and
    # real weight column take the same path — and the weights line up on the
    # index rather than by position.
    w = pd.Series(w, index=c.index, dtype=float)
    # Infinities are not missing values to pandas, so say so before dropping
    # Dropping on the pair drops a household missing EITHER number, which is
    # what keeps the numerator and denominator over the same households.
    df = (pd.concat({'c': c, 'w': w}, axis=1)
          .replace([np.inf, -np.inf], np.nan)
          .dropna())
    # Sum over the poor only. Doing it over everybody looks equivalent, and
    # is — except at alpha=0, where 0**0 == 1 and every household in the
    # country counts as poor.
    poor = df.c < z
    gap = (z - df.c[poor]) / z
    return (df.w[poor] * gap ** alpha).sum() / df.w.sum()

wave = '2016–17'
# food_expenditures is one row per household PER ITEM per source, so it has
# be summed to the household before any of this means anything: without the
# groupby, 'c' is a distribution over individual food purchases, the quartile
# of it is the price of one cheap item, and the headcount counts item-rows
```

```

# rather than people.
c = (cntry.food_expenditures(basis='total').squeeze()
     .xs(wave, level='t').groupby('i').sum())
# fgt() aligns on the index itself, but the mean-weight line below does its
# own boolean indexing, and that needs w and c on identical indexes.
w = sample.xs(wave, level='t').weight.groupby('i').first().reindex(c.index)

z = c.quantile(0.25) # a placeholder line; see the exercise
for a in (0, 1, 2):
    print(f"P_{a}={fgt(c, z, a, w):.4f}")

print(f"unweighted_P_0={fgt(c, z):.4f}")
print(f"mean_weight_poor={w[c < z].mean():.3f}"
      f"non-poor={w[c >= z].mean():.3f}")

```

None of which rescues the line itself. A quartile of the distribution is not a poverty line: it fixes the unweighted headcount at 0.25 by construction, in every country and every year, which is a fact about arithmetic rather than about Ghana. A real line is absolute — the cost of a fixed bundle — and does not move with the distribution, and does not move when everyone gets poorer. Fixing that is the first exercise.

3.3.3 Inequality

- **Lorenz curve:** cumulative share of consumption against cumulative share of population. Everything else is a summary of it.
- **Gini:** twice the area between the Lorenz curve and the 45-degree line. Bounded, familiar, and insensitive where it matters most.
- **Generalized entropy $GE(\alpha)$:** decomposable into within- and between-group parts. $GE(0)$ is mean log deviation, $GE(1)$ is Theil.

3.3.4 Inequality: Atkinson

The opening slide's $I = U(\int w dF) - \int U(w) dF$ is in utils, which nobody reports. Let w_{ede} solve $U(w_{\text{ede}}) = \int U dF$: the level which, given to everyone, yields the same total utility as the actual spread. Then $I = U(\bar{w}) - U(w_{\text{ede}})$, and

$$A = 1 - \frac{w_{\text{ede}}}{\bar{w}}$$

is the same judgement in cedis per cedi, with $U(w) = w^{1-\varepsilon}/(1-\varepsilon)$ and ε assumed, like θ .

$A = 0.3$: this society would give up 30% of mean consumption to have the rest shared equally. Prefer the Lorenz curve to any scalar.

3.3.5 Inequality

The Atkinson index is the opening slide's I with the units fixed. That I is the gap between the utility of the mean and mean utility, which is a quantity in utils, and utils are not a currency anyone reports. Solving for the consumption level that delivers the same mean utility converts the gap into cedis, and dividing by $\bar{w}$ makes it unit free. Same value judgement, expressed as a number a finance ministry can act on.

The judgement is ε , and it is worth being blunt that it is assumed rather than estimated, exactly as θ was in the equivalence scale. Taking $U(w) = w^{1-\varepsilon}/(1-\varepsilon)$,

$$w_{\text{ede}} = \left[\int w^{1-\varepsilon} dF(w) \right]^{1/(1-\varepsilon)},$$

with the limiting case $\varepsilon = 1$ giving the geometric mean, $w_{\text{ede}} = \exp \int \log w dF(w)$. At $\varepsilon = 0$ nobody cares about distribution and $A = 0$ whatever the data look like; as ε grows the index attends more and more to the bottom of the distribution, and in the limit only the poorest household counts. Report it for several values, and let the reader find their own.

Two things follow that are easy to miss. The Atkinson family and the generalized entropy family are the same orderings wearing different clothes: $A(\varepsilon)$ is a monotone transform of $GE(1-\varepsilon)$, so choosing between them is a choice about presentation, not about ethics. And because A is built from a concave U , it is guaranteed to respect the Pigou–Dalton principle, which the Gini also does but which a variance of logs, still common in the applied literature, does not.

3.3.6 Lorenz and Gini

```
def lorenz(c, w):
    c, w = np.asarray(c, float), np.asarray(w, float)
    ok = np.isfinite(c) & np.isfinite(w) & (c >= 0)
    c, w = c[ok], w[ok]
    order = np.argsort(c)
    c, w = c[order], w[order]
```

```

p = np.cumsum(w) / np.sum(w)
L = np.cumsum(w * c) / np.sum(w * c)
return np.concatenate([[0], p]), np.concatenate([[0], L])

def gini(c, w):
    p, L = lorenz(c, w)
    return 1 - 2 * np.trapezoid(L, p)

```

```

print(f"Gini_{food}_purchased_and_own-produced_{wave}={gini(c, w):.3f}")

```

The Atkinson index needs one more thing than the Gini does, and the code should make you say it out loud: how averse to inequality you are.

```

def atkinson(c, w, eps):
    """Weighted Atkinson index.  eps > 0 is inequality aversion."""
    c, w = np.asarray(c, float), np.asarray(w, float)
    ok = np.isfinite(c) & np.isfinite(w) & (c > 0)
    # log/power need c > 0
    c, w = c[ok], w[ok]
    mean = np.sum(w * c) / np.sum(w)
    if np.isclose(eps, 1):
        ede = np.exp(np.sum(w * np.log(c)) / np.sum(w))
    else:
        ede = (np.sum(w * c ** (1 - eps)) / np.sum(w)) ** (1 / (1 - eps))
    return 1 - ede / mean

```

```

for eps in (0.5, 1.0, 2.0):
    print(f"Atkinson_A({eps})={atkinson(c, w, eps):.3f}")

```

The library draws the curve too, and does one thing the code above does not: it writes every choice that moves the curve into the subtitle.

```

import matplotlib.pyplot as plt

```

```

waves = ['1998-99', '2005-06', '2012-13', '2016-17']

```

```

ll.visualizations.lorenz_curve(centry, waves, per='person', basis='total')
plt.show()

```

The Gini in the legend is not the one you computed above, and the subtitle says why: it counts *persons*, weighting each household by its size, where the code above counted households. Both are Ginis of food consumption in

the same year. Neither is wrong; the one you report is a choice, and the chart makes you state it.

If two Lorenz curves cross, the ranking of the two distributions depends on which inequality measure you chose, and no scalar will rescue you. Look before you summarize.

3.3.7 One index, or several?

Everything here divides nominal expenditure by a scalar: a Paasche index over space, a CPI or PPP over time, an equivalence scale over composition. Three scalars, one household, one number out.

- one deflator is exact at **rank one**: budget shares do not vary with total expenditure
- but they do — Engel’s law, which opened this session
- a line *and* a deflator are exact at **rank two** (Gorman), which Engel’s law permits
- so on what grounds is a two-index procedure adequate?

Hold the question: next session answers it for Uganda.

3.3.8 Final thoughts

It would be dishonest to teach the Deaton–Zaidi procedure as best practice and leave it there, because the profession’s own best estimates say the procedure is not adequate for the data we are using it on. It is also the procedure every statistical agency uses, it is what your referees will expect, and you must be able to execute it competently — which is why it comes first and in full.

The order is deliberate: learn the standard method properly here, learn in the next session what it costs, and then decide, case by case, whether the cost matters for the question in front of you. That is a different thing from never having known.

3.3.9 Exercises

1. The poverty line used above is fake. Build a real one: take the food bundle consumed by households in the second and third deciles, price it at national median unit values, and scale it to 2,900 kcal per adult equivalent. Recompute P_0, P_1, P_2 . Notebook: `poverty_line.ipynb`

2. Deflate spatially. Construct a regional Paasche index from the survey's own unit values, apply it, and report how much of the north–south poverty gradient survives. Notebook: `spatial_deflation.ipynb`
3. Vary θ in $c_i = C_i/A_i^\theta$ over $[0.5, 1.0]$ and plot the headcount ratio against it. Over what range does the *ranking* of regions change? Notebook: `equivalence_scales.ipynb`

4 Demand and Welfare

Where household surveys earn their keep for policy: what happens to whom when a price moves. Also: estimated demand functions can help solve a variety of measurement issues.

4.1 Introduction

4.1.1 Reading

Deaton is in your `reading/` folder on the hub, or PDF.

- Deaton, chs. 4 (§4.2 on Engel curves) and 5 (all)
- Deaton & Muellbauer (1980), "An Almost Ideal Demand System," *AER* 70:312–326
- Lewbel (1991), "The rank of demand systems," *Econometrica* 59:711–730
- Ligon, "Estimating Household Welfare from Disaggregate Expenditures" (CUDARE working paper, 2020) — the CFE system we estimate today. In your `reading/` folder on the hub, or eScholarship.

4.2 Engel's Laws and Curves

4.2.1 Ernst Engel

Engel may be the first to use household data in statistical analysis, and used household budget data from Belgium and Saxony to propound what is sometimes called "Engel's (first) Law":

"The greater the income, the smaller the relative percentage of outlay for subsistence."

4.2.2 Budget shares and Engel curves

Let $s_{ij} = p_j q_{ij} / x_i$ be household i 's budget share on good j , x_i total expenditure. The Working–Leser form,

$$s_{ij} = \alpha_j + \beta_j \log x_i + \gamma'_j z_i + u_{ij},$$

fits food shares remarkably well across countries and decades.

- Homothetic utility implies $\beta_{\text{food}} = 0$.
- $\beta_{\text{food}} < 0$ is Engel's law
- z_i : household size and composition, region, season
- Estimated by OLS, clustered on the EA

4.2.3 Estimating the Working–Leser form

Ghana's GLSS7 has both halves of the denominator: food purchases and non-food. Two recall windows, so put both on a per-year basis before adding them. (This is **one** way to do it, and probably not the best).

1. Boring Preface

```
# Show tracebacks without the library's internal frames: the line that
# failed, and why. Change Plain to Verbose if you ever want the rest.
%xmode Plain
# Figures inside the notebook, whatever the kernel's default backend is
%matplotlib inline

# The library audits its own corpus on first read and reports what it
# — implausible quantities, NaN index keys, a column that is wholly
# one wave — at multi-paragraph length. Those reports are a work qu
# whoever maintains the data, not something the room can act on, and th
# the output they are attached to. Silenced here by their own
# "Set LSMS_..._STRICT=1" signature, which is precise: every other warn
# pandas deprecations included, still shows. Delete these two lines to
# them.
import warnings
warnings.filterwarnings("ignore",
                        message=r"(?s).*Set_LSMS_[A-Z_]+=1_to_make_this
```

```
# The first call that builds a table may print "DVC unavailable ...
# falling back to manual aggregation". That is about how the library
# fetches its raw files, not about your data; the numbers are the same,
# and it does not recur once the table is built.
```

```
import matplotlib as mpl
mpl.rcParams.update({'figure.dpi': 110, 'font.size': 11,
                    'lines.linewidth': 1.8})
```

```
import lsms_library as ll
import numpy as np, pandas as pd
```

2. Pull together some data from GLSS7 (2016-17)

```
ghana = ll.Country('GhanaLSS')
wave = '2016-17'
```

```
# The sample table carries the enumeration area v, the weights and the
# rural flag, indexed (i, t). Load each table once; later cells reuse
sample = ghana.sample()
```

```
# market='Region' puts a market level m into the index (and takes v out)
# so the same objects serve the demand-system work later in the session
# waves=[wave] because deriving all seven waves takes three times as long
# and nothing below needs them. GLSS7 asks food at six visits five days
# apart, so summed over items this is thirty days of purchases.
x_f = ghana.food_expenditures(waves=[wave], market='Region')
x_f = x_f.xs(wave, level='t').squeeze().groupby(['i', 'm']).sum()
```

```
# Non-food: Section 9A, one row per item a household reported.
The recall
# window is declared on every row; check it before scaling.
x_n = ghana.nonfood_expenditures(waves=[wave], market='Region').xs(wave)
print(x_n.RecallWindow.unique())
x_n.head()
```

3. Scale some things to improve comparability

```
# One number per household on each side, both per year.
xbar = pd.concat([(x_f * 365.25 / 30).rename('food'),
```

```

x_n.Expenditure.groupby(['i', 'm']).sum().rename('nonfood',
axis=1)

# A household with food but no 9A row at all did not get the last visit
# missing, not zero. Count them, then drop them.
print(f"{xbar.nonfood.isna().sum()} households with food but no 9A row")
xbar = xbar.dropna()
xbar['total'] = xbar.food + xbar.nonfood
xbar['s'] = xbar.food / xbar.total # the food share
xbar.describe().round(2)

```

4. Get household characteristics z The characteristics z are the same as those we'll use later in this session: children and adults by sex, log household size, and a rural dummy. `household_characteristics` counts people by age-sex cell, and `age_cuts` chooses the cells — one cut at 19 gives children and adults; the sample table carries the rural flag.

```

chars = (ghana.household_characteristics(waves=[wave], market='Region')
         .xs(wave, level='t').groupby(['i', 'm']).first())
s1 = sample.xs(wave, level='t') # indexed by i alone
z = chars.join((s1.Rural == 'Rural').astype(float), on='i')

d = pd.concat([xbar.s, np.log(xbar.total).rename('logx'), z], axis=1)
d = d.join(s1.v, on='i').dropna()
d = d[np.isfinite(d.logx)]
len(d)

```

5. Estimate Working-Leser Now the equation on the slide:

$$s_{ij} = \alpha_j + \beta_j \log x_i + \gamma'_j z_i + u_{ij}.$$

Use simple OLS to estimate the coefficients and their covariance matrix. No sampling weights in the regression; they come back when we want population statistics of what it produces.

```

from metrics_miscellany.estimators import ols

zcols = list(z.columns)
X = d[['logx'] + zcols].assign(const=1.0)
b, V = ols(X, d.s)

```

```

coef = b.Coefficients
tab = pd.DataFrame({'coef': coef, 'se': np.sqrt(np.diag(V))})
tab['t'] = tab.coef / tab.se
e = d.s - X @ coef
r2 = 1 - (e**2).sum() / ((d.s - d.s.mean())**2).sum()
print(tab.round(3).to_string())
print(f"\nR^2={r2:.3f}; a doubling of x moves the food share by "
      f"{coef['logx'] * np.log(2):+.3f}")

```

Two things to read off the table.

- (a) β_{food} is negative and well over ten standard errors from zero: **Illustrating Engel's law**
- (b) This rejects homotheticity (which implies $\beta_{\text{food}} = 0$), in one number: a doubling of x lowers the food share by about three points.

Is the line straight? That's what's needed for the Working-Leser or AIDS demand specifications.

Plot the households, the mean share within each expenditure ventile, and the fitted equation with z held at its mean.

```
import matplotlib.pyplot as plt
```

```

d['bin'] = pd.qcut(d.logx, 20, labels=False, duplicates='drop')
curve = d.s.groupby(d.bin).mean()
mid = d.logx.groupby(d.bin).mean()

```

```

zbar = d[zcols].astype(float).mean()
grid = np.linspace(d.logx.quantile(.005), d.logx.quantile(.995), 50)
line = coef['const'] + coef['logx'] * grid + zbar @ coef[zcols]

```

```

fig, ax = plt.subplots(figsize=(6, 4))
ax.scatter(d.logx, d.s, s=3, alpha=0.12, color='0.4', label='households')
ax.plot(mid, curve, 'o', ms=6, label='mean, by ventile')
ax.plot(grid, line, '-', label=r'Working-Leser fit at $\bar{z}$')
ax.set_xlabel(r'$\log x$ (annual expenditure, cedis)')
ax.set_ylabel('food share of expenditure')
ax.set_ylim(0, 1)
ax.legend(frameon=False, loc='lower_left')
plt.show()

```

The ventile means do not follow the line. They sit at 0.75–0.78 across the bottom three-fifths of households and fall only at the top; the R^2 is 0.05. Engel's law is there, but the "remarkable fit" of the slide is not, and the aside on Engel's law as a diagnostic says what a flat share means: something is wrong with the aggregate. Here we can guess what. Section 9A is the *less* frequently purchased non-food. Soap, fuel and transport are asked on the food grid and are not in this table, and rent, utilities, schooling and health are in other sections again. Three-quarters of what a household spends is not food; three-quarters of what we can see is. The `total` understates expenditure and `s` overstates the share, and it does so most for the households whose non-food is mostly the frequent kind (the poor) which is what flattens the curve. The exercises ask what a fuller denominator does to β_{food} .

4.2.4 Engel's Second Law

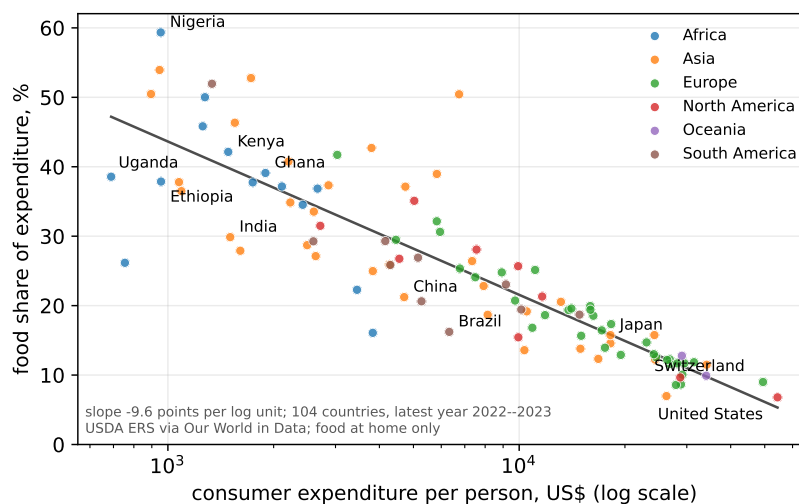
A sort of corollary to the first law, sometimes translated as

"The proportion of the outgo used for food, other things being equal, is the best measure of the material standard of living of a population."

As a welfare measure, this solves some problems that the consumption aggregate has. In particular:

- Is a dimensionless share, so don't need to worry at all about price level, etc.
- If household size/scale can be expressed using Adult Equivalents A_i^θ , then those cancel also.
- Appears to be naturally comparable across time or place.

4.2.5 Engel's law across countries



One point per country, latest year.

4.2.6 Engel's law as a diagnostic

The food share is the oldest welfare indicator in economics and still one of the better ones, precisely because it requires no price data and no deflation. If your constructed consumption aggregate does not produce a declining food share in log total expenditure, something is wrong with the aggregate, and you should find out what before going further. It is the cheapest specification check available, and it catches a surprising share of the errors that arise in session 3 — a mis-scaled own-production imputation, in particular, shows up immediately as a food share that is flat or rising.

There is a serious catch, though: in general while food share can be used to rank the well-being of households *given prices*, if prices change those bets are off as far as comparing across periods. There's basically a different Engel curve for every (p, z) , and needs the structure of a fully estimated demand system to say how changes in prices or characteristics z will change the curve.

4.3 Other Important Elements of Demand Estimation

4.3.1 Equivalence scales from behaviour

Session 3 *assumed* a scale A_i^θ . Demand analysis lets you estimate one.

- **Engel method:** two households are equally well off if they have the same food share. Scale = the expenditure ratio that equalizes it.
- **Rothbarth method:** use expenditure on adult goods (alcohol, tobacco, adult clothing) as the indicator instead.
- The two disagree, systematically and by a lot. Deaton ch. 4 explains why: the Engel method conflates the cost of a child with the child's effect on the household's needs for food specifically.

4.3.2 Prices, and where to get them (or not)

LSMS surveys may not carry reliable data on prices. They carry unit values, and unit values are contaminated by quality choice:

$$\log v_{ij} = \log p_j + \underbrace{\psi_j \log x_i}_{\text{quality}} + \epsilon_{ij}.$$

4.3.3 Use the market comparison

Deaton's solution exploits the design: *within a market*, all households face the same price. So

- regress log unit prices $\log v_{ij}$ on $\log x_i$ and household characteristics *with market fixed effects* to strip out quality;
- the cluster effects are the price variation you wanted.

This is the single best argument for caring about v in the index.

4.3.4 Stripping quality out of unit values

Back in Ghana for a moment, with the objects the Working-Leser cell built. Unit values for the most-reported food item, and Deaton's within-cluster regression of their log on $\log x$:

$$\log v_{ij}^c = a_j^c + \underbrace{\psi_j \log x_i}_{\text{quality}} + \epsilon_{ij}.$$

```
acq = ghana.food_acquired(waves=[wave]).xs(wave, level='t')
uv = np.log((acq.Expenditure / acq.Quantity).replace([np.inf, -np.inf], np.nan))
```

```

j0 = uv.groupby('j').size().idxmax()
u0 = uv.xs(j0, level='j').groupby('i').mean().rename('logv')

e = pd.concat([u0, d.droplevel('m')[['logx', 'v']], axis=1).dropna()

# There are ~1000 clusters, so do not build the dummy matrix; by
# Frisch–Waugh–Lovell, demeaning within cluster gives the identical
# coefficient at a fraction of the cost.
for col in ('logv', 'logx'):
    e[col + '_d'] = e[col].astype(float) - e.groupby('v')[col].transform('mean')

b, V = ols(e[['logx_d']], e.logv_d)
print(f"{j0}: quality elasticity psi = {b.Coefficients['logx_d']:.3f} "
      f"se = {np.sqrt(V.loc['logx_d', 'logx_d']):.3f}")

```

A ψ near zero says households of different means pay the same for this item — it is a homogeneous good. A large ψ says "onion" in the questionnaire is several different goods in the market. Notice that ψ is an *elasticity*, so if we have $\psi = 0.36$ we'd interpret that as evidence that 36% of variation in unit prices was due to different quality choices being made by households with different budgets.

4.3.5 The rank of a demand system

Stack the Engel curves. The *rank* of a demand system (Lewbel 1991) is the dimension of the space spanned by the budget shares viewed as functions of total expenditure.

- rank 1: shares do not vary with x — homothetic
- rank 2: Working–Leser, AIDS
- rank 3: QUAIDS; Gorman's bound for exactly aggregable systems
- rank n : Constant Frisch Elasticity (CFE) (coming up)

Why you should care: **if the rank exceeds one, no single price index can deflate nominal expenditure so as to keep it monotone in utility when relative prices move.** Rank r needs r indices.

4.3.6 The rank of a demand system

This is the result that indicts session 3. Everything we did there — the Deaton–Zaidi household-specific Paasche index, the spatial deflator, the temporal one — constructs a *single* scalar per household and divides by it. That procedure is exactly right if the demand system has rank one, defensible as an approximation at rank two, and simply not available at higher rank: there is no scalar function of prices whose quotient with nominal expenditure orders households the way utility does.

Estimates for Uganda put the rank at three at least, with a best estimate of *six*. So the deflation in session 3 is not a refinement we skipped for lack of time; it is a procedure that cannot be made correct by being more careful with the index. What follows is the alternative: estimate the welfare number directly from behaviour, and never construct a price index at all.

4.4 Constant Frisch Elasticity Demands

4.4.1 Constant Frisch Elasticity demands

Let λ_i be household i 's marginal utility of expenditure. Frischian (λ -constant) demands take the CFE form

$$f_j(p, \lambda) = \left(\frac{\alpha_j}{\lambda p_j} \right)^{\beta_j}, \quad U(c) = \sum_j \alpha_j \beta_j \frac{c_j^{1-1/\beta_j} - 1}{\beta_j - 1}.$$

- β_j is the **Frisch elasticity** of demand for good j : necessities small, luxuries large
- $\beta_j = \beta$ for all j gives CES; $\beta \rightarrow 1$ gives Cobb–Douglas
- unrestricted rank!

4.4.2 The estimating equation

Take logs of expenditure on good j . Writing $w_i \equiv -\log \lambda_i$,

$$\log x_{it}^j = a_t^j + \beta_j w_{it} + \gamma_j' d_{it} + \varepsilon_{it}^j.$$

This is a **one-factor model**. The $(it) \times j$ matrix of log expenditures, once good-time effects a_t^j and characteristics d_{it} are swept out, is rank one: loadings β , factor w .

- estimate by SVD of the residual matrix, with missing data

- d_{it} enters as good-specific *Barten scales*, not "independence of base" adult equivalents.
- needs **no prices** and **no total expenditure** — only expenditures on a subset of goods

4.4.3 Estimating the CFE demand system

The system needs several waves and one food classification that holds across them, coarse enough that most households report most of the goods. Ghana's GLSS has the waves, and since `lsms_library` 0.14.0 it has an `Aggregate Label` as well — but that column folds 210 served labels into 204, which is a renaming rather than an aggregation, and a good only a handful of households ever buy cannot carry a coefficient. Uganda folds 130 into 76. So the session switches countries here, once. The calls are the ones you have already seen, with `age_cuts=(19,)` for the characteristics. The system wants a market index m , and we give it a single national market: the estimator's memory grows with the number of market–wave–good effects it has to sweep out, and the hub gives each of us three gigabytes. Exercise 4 asks what four regional markets change.

```
from cfe import Regression
```

```
# Ghana is done with; the tables it left behind are not small.
del acq, uv, e, x_f, x_n, sample, chars, d
```

```
uga = ll.Country('Uganda')
x = uga.food_expenditures(labels='Aggregate').squeeze()
x = x.groupby(['i', 't', 'j']).sum() # over sources and labels
y = np.log(x.replace(0, np.nan).dropna())
y = pd.concat({1: y}, names=['m']).reorder_levels(['i', 't', 'm', 'j']).sort_index()

z = uga.household_characteristics(age_cuts=(19,)).groupby(['i', 't']).first
z = pd.concat({1: z}, names=['m']).reorder_levels(['i', 't', 'm']).sort_index()

r = Regression(y=y, d=z)
beta = r.get_beta() # a few seconds
print(len(beta), 'goods_estimated')
```

We did not estimate demands for all 76 goods: items too few households ever report cannot support a coefficient, and the estimator drops them without being asked.

```

print("least_elastic:"); print(beta.sort_values().head(6).round(2).to_string)
print("\\nmost_elastic:"); print(beta.sort_values().tail(6).round(2).to_string)
ax = r.graph_beta(xlabel=r'Frisch_elasticities_\\$\\beta_j$')

```

Read those two lists before reading anything else. The ordering is the model's main testable implication and it is not imposed: nothing in the estimator knows which goods are staples. If salt and cassava do not come out at the bottom and fruit and milk at the top, something is wrong with the data or with the aggregation.

Household composition enters through γ , one coefficient per good per characteristic — good-specific Barten scales rather than a single equivalence scale imposed on everything:

```

r.get_gamma().round(2).head(8)

```

Before trusting any of it, look at the fit, drawn the way the Working-Leser figure was: a sample of the observations, the mean of the actual within fifty bins of the predicted, and the line they should lie on. This is also the cell that estimates w — the prediction needs it — so it takes most of a minute.

```

fit = pd.DataFrame({'actual': r.y,
                    'predicted': r.get_predicted_log_expenditures(fill_miss=0)})
fit['bin'] = pd.qcut(fit.predicted, 50, labels=False, duplicates='drop')
mid, mean = fit.groupby('bin').predicted.mean(), fit.groupby('bin').actual.mean()
pts = fit.sample(10000, random_state=0)

fig, ax = plt.subplots(figsize=(5, 5))
ax.scatter(pts.predicted, pts.actual, s=2, alpha=0.1, color='0.4', label='10000 obs')
ax.plot(mid, mean, 'o', ms=4, label='mean_actual_by_bin_of_predicted')
lim = [fit.predicted.quantile(.001), fit.predicted.quantile(.999)]
ax.plot(lim, lim, 'k—', lw=1.2)
ax.set_xlabel('predicted_log_expenditure'); ax.set_ylabel('actual')
ax.set_title(f"R^2={fit.corr().loc['actual', 'predicted']**2:.2f}")
ax.legend(frameon=False, loc='upper_left')
plt.show()

```

4.4.4 The Engel pie

An Engel curve shows one good at a time. With the whole system estimated we can show all of them at once: the predicted budget *composition* as a pie, with the radius growing in $\log x$. Poor households at the centre, rich at the

rim. Under homotheticity every ring would look the same; under Engel's law the staples pinch and the fruit widens.

```

from matplotlib import cm

xhat = r.predicted_expenditures()
xbar = xhat.groupby(['i', 't', 'm']).sum()
p_j = ((r.y.unstack('j') > 0) + 0.).mean()           # prob. good j is bought

lo, hi = xbar.quantile(0.05), xbar.quantile(0.95)
Y = np.geomspace(lo, hi, 60)

fig, ax = plt.subplots(figsize=(7.5, 6))
wedges = None
for k in range(len(Y) - 1, 0, -1):
    ax.set_prop_cycle('color', cm.tab20.colors)
    shares = r.expenditures(Y[k]) * p_j
    radius = 0.3 + 0.7 * np.log(Y[k] / lo) / np.log(hi / lo)
    out = ax.pie(shares, radius=radius, counterclock=False)
    if wedges is None:                               # keep the outermost ring
        wedges, goods = out[0], shares.index.tolist()

big = np.argsort(r.expenditures(Y[-1]).to_numpy() * p_j.to_numpy())[::-1]
ax.legend([wedges[b] for b in big], [goods[b] for b in big],
          loc='center_left', bbox_to_anchor=(1.02, 0.5),
          frameon=False, fontsize=8, title='largest_ten_at_the_rim')
plt.show()

```

Wedges that widen outward are luxuries; wedges that pinch are necessities. This is Engel's law for every good at once, and it is the β plot in a form you can show to someone who has never heard of a Frisch elasticity.

4.4.5 $w = -\log \lambda$ as a welfare measure

The marginal utility of expenditure λ falls as a household gets better off. Hence $w = -\log \lambda$ increases with welfare.

- Comparable across households facing different prices — the prices are in a_t^j , not in w
- Comparable across household compositions — composition is in $\gamma_j' d$

- Derived from behaviour, not from an assumed equivalence scale or an assumed price index

4.4.6 Interpreting the welfare measure

Writing indirect utility as $v(x, p)$, the marginal utility of expenditure is $\lambda = \partial v / \partial x$. Thus $w = -\log \lambda$ is the negative log of that derivative. The elasticity of utility with respect to expenditure would instead be $x\lambda/v$; it is not w .

4.4.7 Welfare comparisons

Two more things come out of the fitted system. The good-time effects a_t^j contain the prices, and `get_pi` separates them into a price index π_t — one number per wave, from expenditures alone — and relative prices. Then `get_w` gives $w = -\log \lambda$ for every household in every wave.

```
pi = r.get_pi()
pi.droplevel('m').round(2)
```

Between 2005-06 and 2009-10 the index jumps by about half a log point: the 2008 food price crisis, found by the demand system with no CPI in sight.

```
w = r.get_w()
w.groupby('t').agg(['size', 'mean', 'std']).round(3)
```

To compare distributions across waves we need a poverty line in units of w . Pin it the way the World Bank's own figures are pinned: PIP puts Uganda's 2005 headcount at 58% (USD 2.15 a day, 2017 PPP), so take the 58th percentile of w in 2005-06 as the line and carry it forward — w is comparable across waves because the prices are in π and a , not in it.

```
from scipy.stats import gaussian_kde
```

```
P0 = 0.58 # PIP headcount, Uganda 2005
zw = w.xs('2005-06', level='t').quantile(P0)
head = w.groupby('t').apply(lambda s: (s < zw).mean())
```

```
waves = sorted(w.index.get_level_values('t').unique())
grid = np.linspace(w.quantile(.005), w.quantile(.995), 300)
fig, ax = plt.subplots(figsize=(6.5, 5))
for k, t in enumerate(waves):
    dens = gaussian_kde(w.xs(t, level='t').dropna())(grid)
    off = (len(waves) - 1 - k) * 0.55
```

```

        ax.fill_between(grid, off, off + dens, color=cm.viridis(k / len(waves))
        ax.plot(grid, off + dens, color='k', lw=0.6)
        ax.text(grid[0], off + 0.05, f'{t}_{head[t]:.0%}', fontsize=9, va='b
ax.axvline(zw, color='r', lw=1)
ax.set_yticks([]); ax.set_xlabel(r'$w_{-}\log\lambda$')
ax.spines[['left', 'top', 'right']].set_visible(False)
plt.show()

```

The whole distribution shifts left into 2009-10 and the headcount rises with it; it recovers by 2010-11, is lowest in 2013-14, and drifts back up after — the figures are on the plot. Nominal expenditure rises in every one of these waves. That comparison, and what it takes to make it honestly, is Thursday's.

Does w mean anything to the households themselves? The Uganda survey asks how many months in the last twelve the household could not meet its food needs — self-reported, discrete, and about as far from a demand system as a welfare question gets. `months_food_inadequate` carries it for 2009-10 to 2015-16 (in 2009-10, 45% of households report at least one such month; in the later waves about 20%). Put the two measures we have beside it, in the last wave that asks the question. Expenditure goes in per adult equivalent on the Oxford scale — 1 for the first adult, 0.7 for each further member aged 14 or more, 0.5 for each child under 14, an assumed scale as session 3 said it would be — and w goes in as it is, because composition is already inside it.

```

mfi = uga.months_food_inadequate() # indexed (i, t, v)
hc = uga.household_characteristics(age_cuts=(14,)).groupby(['i', 't']).first
n14, kids = hc[['F_14+', 'M_14+']].sum(axis=1), hc[['F_00-13', 'M_00-13']]
A = 1 + 0.7 * (n14 - 1).clip(lower=0) + 0.5 * kids # Oxford adult equivalent
q = pd.concat([w.droplevel('m').rename('w'),
               np.log(xbar.droplevel('m') / A).rename('logc')], axis=1)
q = q.reorder_levels(['t', 'i']).join(mfi, how='inner')
q['months'] = pd.cut(q.MonthsInadequate.astype(float),
                    [-0.5, 0.5, 2.5, 5.5, 12.5], labels=['0', '1-2', '3-5'])
qt = q.xs('2009-10', level='t').dropna(subset=['months', 'logc', 'w'])
cats = qt.months.cat.categories

```

```

fig, axes = plt.subplots(1, 2, figsize=(9, 4))
for ax, col, lab in zip(axes, ['logc', 'w'],
                        [r'$\log\lambda_{\text{food expenditure per adult equivalent}}$, r
data = [qt.loc[qt.months == c, col].to_numpy() for c in cats]

```

```

ax.boxplot(data, tick_labels=[f"{c}\n(n={len(v)})" for c, v in zip(cats,
                             showfliers=False, widths=0.6)
ax.set_xlabel('months_without_enough_food_last_12'); ax.set_ylabel(label)
fig.suptitle('Uganda_2009-10: two_welfare_measures_against_what_households_
plt.show()
qt.groupby('months', observed=True)[['logc', 'w']].median().round(2)

```

Both measures tend to fall as the reported months rise. A self-reported measure is a weak instrument, but on this one the two agree with it, and with each other, about who is worse off.

4.4.8 $w = -\log \lambda$ as a welfare measure

What it delivers: a scalar per household-period welfare measure that is comparable across households, estimated from expenditures on a subset of goods, with no price data and no total-expenditure measure. That is a remarkable amount to get from very little.

What it does not deliver: a utility level. One cannot in general integrate back from λ to the indirect utility function — that would require demands for *all* goods. So w supports statements about who is better off than whom, and about how a distribution shifts; it does not support a claim to have measured utility without additional assumptions. Much of the appeal is exactly that the weaker claim is enough for the questions we actually ask of these data.

4.5 Price Changes

4.5.1 Welfare consequences of a price change

How much money makes household i as well off at prices p^1 as it was at p^0 ? With an expenditure function $e(u, p)$,

$$CV_i = e(u_i^0, p^1) - e(u_i^0, p^0), \quad u_i^0 = v(x_i, p^0).$$

4.5.2 A first-order welfare approximation

For a small change in the price of one good j , compensating variation is approximately

$$CV_i \approx q_{ij} \Delta p_j = s_{ij} x_i \frac{\Delta p_j}{p_j}.$$

- to first order you need only the *budget share* — no elasticities

- the distributional incidence follows from how s_{ij} varies with x_i , which is the Engel curve you already estimated
- the exact figure needs the demand system, and says how far the approximation can be trusted

4.5.3 Compensating variation, exactly

The fitted system has an expenditure function, so the exact figure is two calls: indirect utility at the old prices, expenditure at the new ones. Prices come from the system itself — $\pi_t + A_{tj}$ is the log price of good j in wave t — so the experiment is the one Uganda actually ran: 2005-06 prices to 2009-10 prices, for a household with the median 2005-06 food budget and average characteristics.

```
logp = r.get_pi() + r.Ar                                     # (t, m, j): log prices
p05 = np.exp(logp.xs('2005-06', level='t').groupby('j').mean())
p09 = np.exp(logp.xs('2009-10', level='t').groupby('j').mean())

x0 = xbar.xs('2005-06', level='t').median()                 # median food budget
U0 = r.indirect_utility(x0, p05)
cv = lambda p1: float(r.expenditure(U0, p1)) - x0          # e(U0, p1) - e(U0, p0)

print(f"median_2005-06_food_budget: {x0:.0f} shillings")
print(f"CV_of_the_2005-06->2009-10_price_changes: {cv(p09):.0f} shilling")
print(f"({cv(p09)}/{x0:.0%}) of the budget")
```

Two-thirds of the budget: that is what the crisis cost the median household, in the money of the day. Now one good. Matoke has a large Frisch elasticity and its price rose by about 60% between the two waves; raise it alone, and compare the exact answer with the first-order one on the slide.

```
j = 'Matoke'
s_j = r.expenditures(x0, p05)[j] / x0                      # the share the system gives
k_obs = p09[j] / p05[j]
mult = np.linspace(0.5, 3, 26)
exact = np.array([cv(p05.where(p05.index != j, p05[j] * k)) for k in mult])
first = s_j * x0 * (mult - 1)

fig, ax = plt.subplots(figsize=(6, 4))
ax.plot(mult, exact / x0, label='exact: $e(u^0, p^1) - e(u^0, p^0)$')
ax.plot(mult, first / x0, '—', label=r'first_order: $s_j \backslash x \backslash \Delta p_j /$')
```

```

ax.axvline(k_obs, color='0.5', lw=1); ax.axhline(0, color='0.5', lw=1)
ax.set_xlabel(f'price_of_{j},_relative_to_2005-06')
ax.set_ylabel('compensating_variation/_budget')
ax.legend(frameon=False)
plt.show()

```

```

e_obs = cv(p05.where(p05.index != j, p09[j])); f_obs = s_j * x0 * (k_obs -
print(f"at_the_observed_change_(x{k_obs:.2f}):_exact_{e_obs:,.0f},_"
      f"first_order_{f_obs:,.0f};_ratio_{e_obs/_f_obs:.2f}")

```

The first-order line is tangent at the old price and overstates the cost everywhere else, because the household substitutes away and the approximation does not know it. At the change Uganda saw, the overstatement is about a fifth, and it grows with the change. How much that matters depends on the question, which is what the aside is about.

4.5.4 Welfare consequences of a price change

This is the payoff for the whole exercise, and it is worth being explicit about what each answer needs. The first-order answer needs the budget share of the good — by decile, if the question is incidence — which is one `groupby` on a well-constructed consumption aggregate, and no elasticities. A finance ministry contemplating the removal of a fuel subsidy can go that far without a demand system, and should, before arguing about anything else.

But the figure above says how far that goes. At the price change Uganda actually saw, the first-order number overstates the cost by a fifth, because households substitute away and the approximation cannot know it, and the overstatement grows with the change. For all prices at once the first-order term is a price index, and it is the same system that produced one. So the system is not an alternative to the shares. It is where the limits of the shares are measured, and it is what answers the questions the shares cannot: the behavioural response, the revenue from a tax, the welfare comparison across waves in the section before this one.

The order of operations in policy work therefore runs: get the aggregate right, get the shares right, report the first-order incidence with standard errors — and then estimate the system, for what is left.

4.6 Exercises

4.6.1 Exercise 1: compare Engel curves

The Working-Leser cell uses Section 9A alone in its non-food denominator.

1. Interact `logx` with `Rural`. Does the slope differ between urban and rural households?
2. Re-estimate on the 2012-13 wave, which has the same section. Has β moved in five years?

4.6.2 Exercise 1: compare expenditure coverage

Uganda's `nonfood_expenditures()` includes frequently and infrequently purchased non-food items. Put them on a common recall basis:

- The questionnaire specifies 30 days for non-durables and 365 for semi-durables in 2019-20 (see session 3).
- Use `uga.categorical_mapping['nonfood_items']` to identify items.
- Repeat the estimation for Uganda 2019-20. Is the share still flat over most of the distribution? How do the composition coefficients change?

4.6.3 Exercise 1: account for clustered sampling

The reported standard errors allow heteroskedasticity but don't account for dependence within an EA.

- Cluster on the EA, using `d.v.`
- By how much do the standard errors grow?
- Is that the $\sqrt{\text{deff}}$ of session 1?

4.6.4 Exercise 2: equivalence scales

Implement the Engel equivalence scale.

Find the expenditure ratio that equates the predicted food share of:

- a household with two adults and two children;
- a household with two adults.

Compare with the θ -scale from session 2.

4.6.5 Exercise 3: who bears the price change?

For the 2005-06 to 2009-10 price change:

1. Compute the first-order welfare cost at each decile of the 2005-06 food budget, using predicted budget shares.
2. Compute exact compensating variation at the same deciles.
3. Who loses most in shillings? Who loses most as a share of the budget? Are those the same households?

4.6.6 Exercise 4: define regional markets

Re-estimate the CFE system with the four regions as markets. Use `market='Region'` on both tables instead of assigning every household to one national market.

- How much do the β_j move?
- What does π look like by region?
- What did one national market assume away?

Watch your kernel's memory while you do it.

4.6.7 Exercise 5: compare welfare measures

Regress w on log total food expenditure.

- How closely do the two measures agree?
- For which households do they disagree?
- What adjustments does the demand-based measure make that expenditure alone doesn't?

5 From Cross-Sections to Panels

Issues and opportunities that arise from working with longitudinal panels of household data.

5.1 Reading

5.1.1 Reading

Deaton is in your `reading/` folder on the hub, or PDF.

- Deaton, ch. 2 §2.7 and ch. 6
- Mundlak (1961), "Empirical production function free of management bias" *J. of Farm Economics* 43(1):44-56.
- Imbens & Wooldridge (2009), "Linear Panel Data Models," cemap lectures 3–4 (slides I, slides II, notes). Notes sections 1, 3, and 6: fixed effects and first differences, exogeneity, and pseudo-panels.

5.1.2 Reading: cohorts and inference

- Deaton (1985), "Panel data from time series of cross-sections"
- Deaton & Paxson (1994), "Intertemporal Choice and Inequality," *J. of Political Economy* 102(3):437–467 (paper). Section II: following birth cohorts and their consumption dispersion.
- Abadie, Athey, Imbens & Wooldridge (2023), "When should you adjust standard errors for clustering?" *QJE* 138:1–35

5.2 Longitudinal Panels

5.2.1 Introduction to Panels

A panel follows the *same* households across survey waves.

- i identifies a household; t identifies a wave.
- y_{it} is an outcome; z_{it} is a vector of observed covariates.
- Changes within a household let us hold its persistent characteristics fixed, even when we cannot measure them.
- Repeated cross-sections follow a population; a panel also follows its members. Matching household identifiers is what makes the difference.

5.2.2 Panels in lsms_library

The library's `panel_ids` crosswalks connect wave-specific identifiers. The returned index level `i` uses those links; matching raw household numbers ourselves can join unrelated households.

First find the surveys with a declared crosswalk. This reads the catalogue, without loading their household records.

```
import warnings
warnings.filterwarnings("ignore",
                        message=r"(?s).*Set_LSMS_[A-Z_]"
                        "+=1_to_make_this_fatal")

import lsms_library as ll
import numpy as np
import pandas as pd
from IPython.display import display

panel_countries = {}
for name in ll.countries():
    country = ll.Country(name)
    if 'panel_ids' in country.data_scheme:
        panel_countries[name] = country
print(", ".join(panel_countries))
```

The following snapshot uses extracts checked on 16 September 2026. *Waves* counts the survey's available `t` labels, and the last two columns give the smallest and largest *whole-wave* household counts in `sample()`. They are not the size of a balanced panel.

Nigeria's ten labels are planting/harvest visits in five survey waves. Only GhanaLSS's first two waves are linked. Ethiopia and Niger contain separate panel cohorts; Tanzania has extended and refreshment panels. A country's full list of waves is not one continuously followed sample.

Recompute the table from your installed library. We count distinct households, not rows of an expenditure or person-level table.

```
inventory_rows = []
for name, country in panel_countries.items():
    sample_counts = (country.sample().reset_index()
                     .groupby('t')['i'].nunique())
    inventory_rows.append({
        'Survey': name,
        'Waves_listed': len(country.waves),
        'Waves_in_sample': len(sample_counts),
        'Minimum_households': sample_counts.min(),
```

Table 1: Surveys declaring household crosswalks; sample-table coverage.

Survey	Waves	Minimum households	Maximum households
Burkina_Faso	3	3,227	10,411
Ethiopia	5	3,969	6,770
EthiopiaRHS	8	338	1,480
GhanaLSS	7	3,147	16,772
Malawi	5	4,000	14,955
Mali	4	3,804	8,390
Niger	4	3,558	6,622
Nigeria	10	4,696	5,263
Senegal	2	7,120	7,156
Tanzania	6	1,184	5,010
Uganda	8	2,716	3,305

```

        'Maximum_households': sample_counts.max(),
    })
panel_inventory = pd.DataFrame(inventory_rows).set_index(
    'Survey')
display(panel_inventory)

```

The catalogue also contains these documented panel components without a declared `panel_ids` crosswalk. They require further linkage work before the same exercise is justified; the absence of a library crosswalk does not mean the survey was a repeated cross-section.

Table 2: Panel components documented in the library’s country notes.

Survey	Panel component
Albania	2002–04; three waves
CotedIvoire	1985–88; four rounds, rotating two-year panels
GhanaSPS	2009-10, 2013-14, 2017-18; three waves
Nepal	1995-96, 2003-04, 2010-11; panel subsamples
Peru	Partial links 1985–90 and 1991–94; coverage varies
Serbia and Montenegro	2002–03; two waves
Tajikistan	2007–09; two waves

5.2.3 Two-way Fixed Effects Regressions

Consider a common-slope model:

$$y_{it} = \alpha_i + \delta_t + \beta' z_{it} + \epsilon_{it}.$$

- α_i : persistent household differences, such as ability or land quality. These may be correlated with z_{it} .
- δ_t : shocks shared by every household in a wave.
- β : identified by changes in z_{it} within households, after removing common time movements.

A time-invariant covariate is absorbed by α_i ; a covariate that varies only by wave is absorbed by δ_t .

5.2.4 TWFE generalizes a double difference

Two groups, A and B ; two dates, before (0) and after (1). Apply the TWFE model to their mean outcomes, with treatment only in $(A, 1)$. Its treatment coefficient is

$$\hat{\beta} = (\bar{y}_{A1} - \bar{y}_{A0}) - (\bar{y}_{B1} - \bar{y}_{B0}).$$

- Difference over time: each group's fixed effect cancels.
- Difference across groups: the common time change cancels.
- TWFE extends this logic to more units, dates, and covariates.

A causal interpretation needs parallel *untreated* trends and no anticipation. More complicated designs need further care.

5.2.5 What makes the within comparison valid?

A sufficient *strict exogeneity* condition is

$$\mathbb{E}[\epsilon_{it} \mid z_{i1}, \dots, z_{iT}, \alpha_i] = 0 \quad \text{for every } t.$$

- Covariates at *all* dates must be unrelated to the idiosyncratic shock at t . Feedback into future covariates can violate this.
- After removing both effects, the covariates must still have full column rank.
- Large-sample arguments require many independent sampling units and suitable moment conditions.

Allow for serial correlation within households and dependence within sampling clusters when computing standard errors.

5.2.6 What fixed effects can remove

- A time-constant omitted household characteristic: absorbed by α_i , if it enters additively.
- A common additive time shock: absorbed by δ_t .
- A persistent additive reporting error: also absorbed, if its contribution to the equation stays constant.

What remains a problem:

- household-specific shocks correlated with changing covariates;
- time-varying effects of an omitted characteristic;
- time-varying measurement error in the regressors.

Within variation can contain proportionally *more* measurement error.

5.2.7 What the reporting-error qualification means

Suppose the true covariate is z_{it}^* , but we observe $z_{it} = z_{it}^* + a_i$. In a linear common-slope equation, $-\beta' a_i$ becomes part of the household intercept. If instead the reporting error changes with t , demeaning leaves it in the regressor and the regression error. Classical regressor measurement error is therefore not generally repaired by fixed effects.

Mean-zero measurement error in the *outcome* need not bias the slope when it is unrelated to the covariates, although it reduces precision. These are different problems.

5.2.8 Attrition

Attrition is loss of intended follow-up observations. It is not every decline in the number of household IDs shared by two tables.

- Planned rotation removes households by design.
- Refreshment adds newly sampled households.
- Splits change the household being followed.
- Missing modules can remove observations from a regression even when the household was interviewed.

Read the sampling and tracking rules before calling an overlap loss an attrition rate.

5.2.9 Attrition and consistency

Let $R_{it} = 1$ when the observation is available, and let $R_i = (R_{i1}, \dots, R_{iT})$. A sufficient condition for the common-slope model is

$$\mathbb{E}[\epsilon_{it} \mid z_{i1}, \dots, z_{iT}, \alpha_i, R_i] = 0.$$

- Selection may depend on α_i and on the covariates.
- Selection driven by idiosyncratic outcome shocks can violate this condition, even if the shock precedes departure.
- We still need identifying variation in the observed sample.

Keeping only households observed in every wave does not repair selection on shocks.

5.2.10 Population changes after attrition

Consistency of a common regression slope is different from representativeness of a mean or a poverty rate. With heterogeneous slopes, changes in who remains can also change which effects the regression averages.

Compare stayers and leavers on baseline observables, but agreement cannot establish selection independent of unobserved shocks. Inverse-probability response weights require a defensible response model and positive response probabilities. Ordinary cross-sectional survey weights need not supply that correction.

See Imbens and Wooldridge’s lecture 18, section 1, pp. 4–5, for fixed effects conditional on the observation pattern, and section 2 for weighting.

5.2.11 A genuine panel: Uganda

The Uganda National Panel Survey draws its initial households from the 2005–06 household survey. The library carries eight time labels through 2019–20.

- Continuing households identify within-household changes.
- Split-offs and refreshment households also enter the data.
- Sample refreshment began in 2013–14; a decline in overlap across that boundary includes a design change.

We first examine 2009–10, 2010–11, and 2011–12, before that refresh. UBOS sampling and tracking description.

5.2.12 Observed Attrition in the Uganda Panel

Compare the sample table with roster-derived household characteristics. An absent characteristics record need not mean the household is absent from the sample table.

```
uga = ll.Country('Uganda')
uga_waves = ['2009-10', '2010-11', '2011-12']
uga_sample = uga.sample(waves=uga_waves)
uga_chars = uga.household_characteristics(waves=uga_waves)
uga_tables = {'sample': uga_sample, 'characteristics':
    uga_chars}
```

`Country.panel_attrition()` uses the roster. For a particular dataframe, call `ll.tools.panel_attrition`. The diagonal is the number of households in that wave; a cell above it counts IDs present in *both* waves. It does not require presence in every intervening wave.

```
uga_overlap = {}
for name, frame in uga_tables.items():
    observed_waves = sorted(frame.index.get_level_values(
        't').unique())
    assert observed_waves == uga_waves, (name,
        observed_waves)
    counts = ll.tools.panel_attrition(
        frame, observed_waves,
        split_households_new_sample=True)
    uga_overlap[name] = counts
    print(name)
    display(counts.astype('Int64'))
```

For a common starting date, express later overlap as a percentage of the first diagonal entry. We leave the matrix's unused lower triangle missing rather than displaying it as zero.

```
uga_overlap_pct = pd.DataFrame({
    name: 100 * counts.loc[uga_waves[0]].astype(float)
        / float(counts.loc[uga_waves[0], uga_waves[0]])
    for name, counts in uga_overlap.items()
})
display(uga_overlap_pct.round(1))
```

In the checked extracts, both tables start with 2,975 households. The sample table shares 2,587 of those IDs with 2011–12; the characteristics table shares 2,582. Read the matrix for the variables you will actually use, not just for the survey as a whole.

The default counts exact IDs. With `split_households_new_sample=False` the library also counts later descendant households. A parent can have several descendants, so that alternative count divided by the initial sample size need not be a retention probability.

5.2.13 Removing Fixed Effects: a balanced panel

A *balanced* panel observes every household in the same T waves. An *unbalanced* panel observes different subsets of waves.

In a balanced, equally weighted panel, define for any variable q

$$\tilde{q}_{it} = q_{it} - \bar{q}_{i\cdot} - \bar{q}_{\cdot t} + \bar{q}_{\cdot\cdot}$$

Apply this transformation to y and *every* covariate:

$$\tilde{y}_{it} = \beta' \tilde{z}_{it} + \tilde{\epsilon}_{it}.$$

Both sets of intercepts disappear. Regress the transformed variables without adding an intercept.

5.2.14 Removing Fixed Effects: an unbalanced panel

Household demeaning still removes α_i , using each household's observed waves. Subtracting the two sets of means *once* generally does not remove both effects.

Let D contain household and time indicators on the observed rows:

$$M_D = I - D(D'D)^+ D', \quad \hat{\beta} = (Z' M_D Z)^{-1} Z' M_D y.$$

This is Frisch–Waugh–Lovell:

1. Remove the fitted values from projecting y and each column of Z on the *same* D .
2. Regress the residual outcome on the residual covariates.

5.2.15 Computing the projection

The superscript $+$ denotes a generalized inverse; alternatively, drop redundant indicator columns. The formula describes a projection, not an instruction to allocate an observation-by-observation matrix.

One exact implementation first subtracts household means from the outcome, covariates, *and time indicators*. Regress the demeaned outcome on the

demeaned covariates and demeaned time indicators. Applying FWL again to those time indicators gives the displayed estimator. Demeaning the covariates but leaving the time indicators unchanged does not give the same regression.

A household seen once contributes zero to this within regression. The algebra delivers the same slope as fitting the full indicator model; inference must still account for dependence and for any degrees-of-freedom adjustment associated with absorbed effects. The weighted problem uses weighted projections.

5.2.16 A GLSS panel

GLSS1 (1987–88) and GLSS2 (1988–89) contain a two-wave panel.

- Half of GLSS1’s sampling workloads were retained for GLSS2; half were replaced.
- `PANELC.DAT` provides person-level links. The library applies these, keeping split-off households distinct.
- Later GLSS rounds have no registered household linkage.
- Loss of overlap includes planned rotation, not just missed follow-up interviews.

GSS, *GLSS3 Report*, Appendix 1, p. 110.

5.2.17 Which weights for the GLSS panel?

The first two *cross-sections* were self-weighting. The linked, responding subsample need not be.

For a baseline-population mean, a panel weight would account for

$$w_i^P \propto \frac{w_i^0}{\Pr(\text{retained} \mid i \text{ sampled}) \Pr(\text{responds} \mid i \text{ retained})}.$$

- The library’s `GLSS1/2 weight` and `panel_weight` are both one. The latter is not a documented correction for panel response.
- We report unweighted descriptions of the observed linked sample. These are not population estimates of change.

5.2.18 Examine the GLSS Panel

Start with the full sample in the two relevant waves, and let the library return both its overlap matrix and the linked identifiers. Do this *before* dropping households, so the selection remains visible.

```
ghana = ll.Country('GhanaLSS')
glss_waves = ['1987-88', '1988-89']
glss_sample = ghana.sample(waves=glss_waves)
glss_overlap, glss_ids = ll.tools.panel_attrition(
    glss_sample, glss_waves, return_ids=True,
    split_households_new_sample=True)
display(glss_overlap.astype('Int64'))

glss_linked_ids = glss_ids[tuple(glss_waves)]
glss_sample_panel = glss_sample[
    glss_sample.index.get_level_values('i').isin(
        glss_linked_ids)]
assert glss_sample_panel.groupby(level='i').size().eq(2).
    all()
print(f"{len(glss_linked_ids):,} households linked in
    both sample tables")
display(glss_sample_panel.groupby('t')[['weight', '
    panel_weight']]
    .agg(['count', 'min', 'max']))
```

The checked sample tables contain 3,147 and 3,194 households, of which 712 share an ID. This is smaller than the 714 distinct GLSS2 households in the raw person crosswalk: descendants with their own IDs are not the same unit as their parent. Do not collapse those IDs to force a larger panel.

Next compare availability in two substantive tables. A roster can give us household size even when no food-expenditure row is available.

```
glss_chars = ghana.household_characteristics(waves=
    glss_waves)
glss_food = ghana.food_expenditures(waves=glss_waves)
for name, frame in [('characteristics', glss_chars), ('
    food', glss_food)]:
    print(name)
    display(ll.tools.panel_attrition(frame, glss_waves).
        astype('Int64'))
```

Construct one row per household and wave. Household size comes from `log HSize`; summing all characteristics columns would incorrectly add that logarithm to the age-sex counts. Food value sums items and sources, preserving an entirely missing sum as missing.

```

glss_size = (np.exp(glss_chars['log_HSize']).groupby(['t',
    , 'i']))
    .first().rename('household_size'))
glss_food_value = (glss_food['Expenditure'].groupby(['t',
    'i']))
    .sum(min_count=1).rename('food_value')
glss_values = pd.concat([glss_size, glss_food_value],
    axis=1)
glss_values = glss_values[
    glss_values.index.get_level_values('i').isin(
        glss_linked_ids)]
display(glss_values.groupby('t').count())

```

Finally require both variables in both waves. Report the number lost at this step, then describe the *same households* in each wave.

```

glss_complete = glss_values.dropna()
glss_complete = glss_complete[np.isfinite(glss_complete).
    all(axis=1)]
observations = glss_complete.groupby(level='i').size()
glss_complete_ids = observations.index[observations.eq(
    len(glss_waves))]
glss_panel = glss_complete[
    glss_complete.index.get_level_values('i').isin(
        glss_complete_ids)]
assert glss_panel.index.is_unique
assert len(glss_panel) == len(glss_waves) * len(
    glss_complete_ids)
assert len(glss_complete_ids) > 0, "No complete linked
    households"

print(f"Linked sample households: {len(glss_linked_ids)}
    :}")
print(f"Complete two-wave households: {len(
    glss_complete_ids):}")
print(f"Lost for missing variables: "
    f"{len(glss_linked_ids) - len(glss_complete_ids):}")
display(glss_panel.groupby('t').agg(['count', 'mean', '
    median', 'std'])
    .round(2))
display(ll.tools.panel_attrition(glss_panel, glss_waves).
    astype('Int64'))

```

In the checked extracts, 706 of the 712 linked sample households have

both variables in both waves. Their unweighted means are:

Table 3: Complete two-wave GLSS sample, extracts checked 16 September 2026.

Wave	Households	Household size	Recorded food value
1987-88	706	5.33	4,545.55
1988-89	706	5.50	5,202.56

Balancing the rows makes the comparison well defined; it does not establish that the households who remain represent those who leave.

Food value is in nominal survey currency, using the library’s food valuation and recall conventions. A change in this number is not a change in real consumption or welfare.

5.3 Application: Farm Management in Ghana

5.3.1 Management and measured productivity

Farms with more inputs often produce more. They may also be managed differently.

- A farmer who chooses better planting dates may also cultivate more land or hire more labor.
- A cross-sectional regression can attribute that persistent advantage to the measured inputs.
- Repeated observations let us compare a farm with itself, holding fixed an additive management effect.

Mundlak (1961) motivates this use of farm effects. We apply the idea to household crop values in the two-wave GLSS panel.

5.3.2 Cobb–Douglas: technology and management

For one output and positive inputs, suppose

$$Q_{it} = A_{it} H_{it}^{\beta_H} L_{it}^{\beta_L} K_{it}^{\beta_K}.$$

- H : land; L : labor; K : another productive input.
- A : productivity, including management and unmeasured conditions.

- Each β is a physical-output elasticity in this model.
- Scaling all inputs scales output by the factor raised to $\beta_H + \beta_L + \beta_K$.

The form describes a technology. It doesn't say inputs are chosen independently of productivity.

5.3.3 From output to a linear revenue equation

If revenue is $R_{it} = p_{it}Q_{it}$, taking logs gives

$$\log R_{it} = \beta_H \log H_{it} + \beta_L \log L_{it} + \underbrace{\log p_{it} + \log A_{it} + \beta_K \log K_{it}}_{u_{it}}.$$

- An unmeasured input joins productivity and price in the residual.
- Correlation between that residual and measured inputs biases OLS.
- A sum of crop values needn't have this single-output structure.

Our Ghana regression is a log-linear description of *partial crop value*. Its coefficients aren't automatically physical-output elasticities.

5.3.4 What the Ghana tables observe

- Output: values sold, kept for seed, given away, and lost; own consumption is absent.
- Land: harvested area summed across crops.
- Labor: family hours, hired expenditure, exchange days.
- Inputs: recorded spending, with marketing costs separate.

Harvest quantities are unavailable. Valuations mix sale and replacement prices. We won't infer missing output from land.

5.3.5 Construct the household-year observations

We use the two waves supplied by LSMS_Library PR #934. The feature branch is `analysis/panel-estimation`, checked at `8394afc54`.

The construction follows `HOWTO_MUNDLAK.org`. A release without the Ghana agricultural tables cannot run it.

Each table has more detail than our household regression needs. Aggregate by the named `t` and `i` levels, retaining a wholly missing sum as missing. This also retains records whose cluster identifier is missing.

```

import warnings
warnings.filterwarnings("ignore",
                        message=r"(?s).*Set_LSMS_[A-Z_]+=1_to_make_this_fatal")

import numpy as np
import pandas as pd
import datamat as dm
import lsms_library as ll
from scipy.stats import chi2
from IPython.display import display

farm_country = ll.Country('GhanaLSS')
farm_waves = ['1987-88', '1988-89']
farm_features = ['crop_production', 'labor', 'plot_labor',
                 , 'plot_inputs']
assert all(name in farm_country.data_scheme for name in
           farm_features), (
    "Use the Ghana agricultural feature branch from PR
    #934")
print("Library source:", ll.__file__)

def farm_sum(series, name):
    return series.groupby(level=['t', 'i']).sum(min_count
        =1).rename(name)

farm_crop = farm_country.crop_production(waves=farm_waves
    )
farm_jobs = farm_country.labor(waves=farm_waves)
farm_work = farm_country.plot_labor(waves=farm_waves)
farm_inputs = farm_country.plot_inputs(waves=farm_waves)
farm_sample = farm_country.sample(waves=farm_waves)

```

First construct the dependent variable. Adding the four dispositions produces a partial value, in nominal pre-2007 cedis. It mixes sale receipts with replacement valuations and omits food consumed by the household. The default API does not deflate these amounts. A common multiplicative price change can enter a wave effect; household-specific prices cannot.

A partially observed sum remains partial. The coverage table counts positive-area crop rows with no observed disposition value at all.

```

farm_value_columns = ['Value_sold', 'Value_seed', '
    Value_given', 'Value_lost']
farm_crop_value = farm_crop[farm_value_columns].sum(axis
    =1, min_count=1)
farm_output = farm_sum(farm_crop_value, 'partial_value')

```

```

farm_area = farm_sum(farm_crop['Area_ha'], 'crop_area_ha'
)
positive_area = farm_crop['Area_ha'].gt(0)
coverage = pd.DataFrame({
    'positive_area_rows': positive_area,
    'no_value_rows': positive_area & farm_crop_value.isna
    (),
}).groupby('t').sum()
coverage['no_value_percent'] = (100 * coverage['
no_value_rows']
                                / coverage['
positive_area_rows'])
display(coverage.round(1))

```

5.3.6 Put the three labor measures in common units

Family labor is annual weeks times usual hours per week. The industry codes 100–199 are the HOWTO’s working agriculture classification, not a verified crop-only classification. They can include forestry and fishing. The library combines annual information from the reference-week job modules with supplementary annual jobs; repeated-job pointer rows carry no additional hours and mustn’t create extra labor.

For hired labor, divide expenditure by the median *day* wage of paid agricultural workers in that wave. For family labor, divide hours by the pooled median reported hours per day. Display the observations behind both conversions. These are approximations to comparable labor services, not measurements of identical workers or tasks.

```

farm_agri = farm_jobs['Industry'].between(100, 199).
    fillna(False).astype(bool)
farm_paid = (farm_agri & farm_jobs['OwnFarmOrBusiness'].
    eq(False).fillna(False).astype(bool)
    & farm_jobs['Earnings'].gt(0).fillna(False).
    astype(bool)
    & farm_jobs['EarningsUnit'].eq('Day').fillna
    (False).astype(bool))
farm_wage_summary = (farm_jobs.loc[farm_paid, 'Earnings'
    ].groupby('t')
    .agg(['count', 'median']))
farm_day_wage = farm_wage_summary['median']
farm_hours = farm_jobs.loc[farm_agri, 'HoursPerDay'].
    dropna()
farm_hours_per_day = farm_hours.median()
assert farm_day_wage.reindex(farm_waves).gt(0).all()

```

```

assert farm_hours_per_day > 0
print("Hours/day_observations:", len(farm_hours),
      ";_pooled_median:", farm_hours_per_day)
display(farm_wage_summary)

farm_annual_hours = farm_jobs['WeeksPerYear'] * farm_jobs
    ['HoursPerWeek']
farm_own = farm_agri & farm_jobs['OwnFarmOrBusiness'].eq(
    True).fillna(False).astype(bool)
farm_family_days = (farm_sum(farm_annual_hours.where(
    farm_own), 'family_days')
    / farm_hours_per_day)
farm_source = farm_work.index.get_level_values('source')
farm_hired_cost = farm_sum(
    farm_work['Cost'].where(farm_source == 'hired'), '
    hired_cost')
farm_exchange = farm_sum(
    farm_work['PersonDays'].where(farm_source == '
    exchange'), 'exchange_days')
farm_data = pd.concat([farm_output, farm_area,
    farm_family_days,
                        farm_hired_cost, farm_exchange],
                        axis=1)
farm_data['hired_days'] = (farm_data['hired_cost']
    / farm_data.index.get_level_values('t').map(
    farm_day_wage))

```

The HOWTO sums available family, hired, and exchange days, treating absent components as zero only when at least one component is observed. The library drops recorded zero hired/exchange cells, but absence alone can also reflect missing information. We retain the component-availability counts so this assumption is visible. It can understate labor where an unobserved component was positive.

The country notes identify three marketing costs: transport, containers, and storage. Keep them separate from purchased production inputs. Neither input cost nor hired cost needs to be positive for our main land-and-total-labor specification.

```

farm_parts = ['family_days', 'hired_days', 'exchange_days
    ']
farm_data['labor_components_observed'] = farm_data[
    farm_parts].notna().sum(axis=1)
farm_data['labor_days'] = farm_data[farm_parts].sum(axis
    =1, min_count=1)
display(pd.crosstab(farm_data.index.get_level_values('t')

```

```

        ,
        farm_data['labor_components_observed'
        ]))
farm_marketing = farm_inputs.index.get_level_values('
input').isin(
    ['Transport', 'Containers', 'Storage'])
farm_data['production_input_cost'] = farm_sum(
    farm_inputs['Cost'].where(~farm_marketing), '
    production_input_cost')
farm_data['marketing_cost'] = farm_sum(
    farm_inputs['Cost'].where(farm_marketing), '
    marketing_cost')
display(farm_data.groupby('t')[farm_parts + ['labor_days'
]].median().round(1))

```

5.3.7 Define the estimation sample

Start from linked sample IDs, preserving the library's split-household convention. Don't reuse `glss_panel` from the food example: its food-availability restriction is irrelevant here.

Show overlap in the actual tables, then require finite, positive partial value, crop area, and total labor in *both* waves. A missing disposition isn't evidence of zero production. Taking logs and balancing select a sample; neither operation repairs selection bias.

```

farm_overlap, farm_links = ll.tools.panel_attrition(
    farm_sample, farm_waves, return_ids=True,
    split_households_new_sample=True)
farm_linked_ids = farm_links[tuple(farm_waves)]
for name, table in [('sample', farm_sample), ('crops',
    farm_crop),
                    ('annual_labor', farm_jobs), ('hired/
    exchange', farm_work)]:
    print(name)
    display(ll.tools.panel_attrition(table, farm_waves).
        astype('Int64'))

farm_columns = ['partial_value', 'crop_area_ha', '
    labor_days']
farm_usable = farm_data[farm_columns].replace([np.inf, -
    np.inf], np.nan).dropna()
farm_usable = farm_usable[farm_usable.gt(0).all(axis=1)]
print("Usable_household-years_before_linkage/balancing:",
    len(farm_usable))

```

```

farm_candidates = farm_usable[
    farm_usable.index.get_level_values('i').isin(
        farm_linked_ids)]
farm_nwaves = farm_candidates.groupby('i').size()
farm_ids = farm_nwaves.index[farm_nwaves.eq(2)]
farm_panel = farm_candidates[
    farm_candidates.index.get_level_values('i').isin(
        farm_ids)].sort_index()
assert farm_panel.index.is_unique
assert len(farm_panel) == 2 * len(farm_ids) and len(
    farm_ids) > 0
farm_log = np.log(farm_panel.astype(float))
print("Linked_sample_households:", len(farm_linked_ids))
print("Complete_farm_households:", len(farm_ids), ";_
    panel_rows:", len(farm_panel))
print("Lost_from_linked_sample:", len(farm_linked_ids) -
    len(farm_ids))
display(farm_panel.groupby('t').agg(['count', 'mean', '
    median']).round(2))
farm_input_spend = farm_data.loc[farm_panel.index, '
    production_input_cost']
display(pd.DataFrame({'missing': farm_input_spend.isna(),
    'zero': farm_input_spend.eq(0),
    'positive': farm_input_spend.gt(0)
    }).groupby('t').sum())
print("Farms_with_positive_production-input_spend_in_both
    waves:",
    int(farm_input_spend.gt(0).fillna(False).groupby('i')
        .all().sum()))

```

This additional positivity restriction would select a different farm sample. Its count needn't match the HOWTO's purchased-input example, which sums marketing and production costs together.

5.3.8 Diagnose land before estimating

The land regressor sums harvested area across crops. Multiple cropping and intercropping can make this different from physical land operated. Changes in crop coverage or reporting can create changes in this sum.

Count positive-area crop records and compare crop sets across the two waves. The diagnostic below reports the root mean squared within-farm log-area deviation on the full panel and on farms with the same crop set. It uses the same definition as the HOWTO's within standard deviation. A stable crop set doesn't establish accurate area, and restricting to it selects farms

partly on a production choice.

```

farm_grown = farm_crop.loc[farm_crop['Area_ha'].gt(0)].
    reset_index()
farm_crop_sets = farm_grown.groupby(['t', 'i'])['j'].agg(
    frozenset)
farm_crop_counts = farm_crop_sets.map(len)
display(farm_crop_counts.groupby('t').describe().round(2)
)
farm_sets_wide = farm_crop_sets.unstack('t').reindex(
    farm_ids)
farm_same_crops = farm_sets_wide[farm_waves[0]].eq(
    farm_sets_wide[farm_waves[1]])
farm_stable_ids = farm_same_crops.index[farm_same_crops]
farm_land_diagnostics = []
for name, ids in [('all_panel_farms', farm_ids),
    ('same_crop_set', farm_stable_ids)]:
    log_area = farm_log.loc[farm_log.index.
        get_level_values('i').isin(ids),
        'crop_area_ha']
    within_area = log_area - log_area.groupby('i').
        transform('mean')
    farm_land_diagnostics.append({
        'sample': name, 'farms': len(ids),
        'within_log_area_rms': np.sqrt(within_area.pow(2)
            .mean())})
display(pd.DataFrame(farm_land_diagnostics).set_index('
    sample').round(3))
farm_area_change = (farm_log['crop_area_ha'].unstack('t')
    [farm_waves[1]]
        - farm_log['crop_area_ha'].unstack('t'
            ') [farm_waves[0]])
display(farm_area_change.describe(percentiles=[.1, .5,
    .9]).round(3))

```

At the checked feature revision, the within log-area RMS is 1.117 for all 506 farms and 1.073 for the 21 farms with unchanged crop sets. The median log-area change is 1.924. This diagnostic doesn't resolve the area problem: holding the reported crop set fixed leaves large changes, and does so on a very small, selected subsample.

5.3.9 Cross-sectional estimation

At a single date, regress log crop value on log area and log labor:

$$y_i = c + \beta_H h_i + \beta_L \ell_i + u_i.$$

- Management, omitted inputs, and prices enter u_i .
- OLS needs this residual uncorrelated with the regressors.
- Productive farmers may choose more inputs: coefficients can combine input effects with selection across farms.

Compare each wave with changes among the *same* households.

5.3.10 Estimate with DataMat

`DataMat.lstsq` solves least squares. It doesn't choose an intercept, align two unrelated samples, or supply clustered uncertainty for us. We provide the same ordered rows of outcome and regressors and check rank before fitting.

For the later farm-mean test, add a farm-cluster sandwich covariance. Sum the regressor-times-residual scores within each farm, then form

$$\hat{V} = \frac{G}{G-1} \frac{n-1}{n-k} (X'X)^{-1} \left(\sum_i s_i s_i' \right) (X'X)^{-1}.$$

Here G counts farms, n rows, and k fitted coefficients. This CR1 covariance permits dependence between a farm's two observations. It assumes independent farms; dependence across farms in a sampling cluster would require clustering at that level instead.

The estimator is not about farms, so we give it a neutral name and use it again on the Uganda panel at the end of the session, where the cluster is the household. `coef_table` prints a fit the way a regression table reads.

```
def clustered_ols(X, y):
    assert X.index.equals(y.index)
    X, y = X.astype(float), y.astype(float)
    n, k = X.shape
    assert np.isfinite(X.to_numpy()).all() and np.
        isfinite(y).all()
    assert np.linalg.matrix_rank(X.to_numpy()) == k and n
        > k
    b = pd.Series(np.asarray(dm.DataMat(X).lstsq(dm.
        DataVec(y))).reshape(-1),
        index=X.columns)
    residual = y - X @ b
    scores = X.mul(residual, axis=0).groupby(level='i').
        sum()
    G = len(scores)
    assert G > 1
```

```

bread = np.linalg.inv(X.to_numpy().T @ X.to_numpy())
V = bread @ (scores.to_numpy().T @ scores.to_numpy())
    @ bread
V *= (G / (G - 1)) * ((n - 1) / (n - k))
return b, pd.DataFrame(V, index=X.columns, columns=X.
    columns)

def coef_table(b, V):
    """Coefficients, cluster-robust standard errors, and
    t statistics."""
    se = pd.Series(np.sqrt(np.diag(V)), index=b.index)
    return pd.DataFrame({'coef': b, 'std_err': se, 't': b
        / se}).round(3)

farm_xcols = ['crop_area_ha', 'labor_days']
farm_cross = {}
for wave in farm_waves:
    wave_data = farm_log.xs(wave, level='t').sort_index()
    X = wave_data[farm_xcols].assign(const=1.0)
    farm_cross[wave] = clustered_ols(X, wave_data['
        partial_value'])
farm_pooled_X = farm_log[farm_xcols].assign(
    second_wave=(farm_log.index.get_level_values('t') ==
        farm_waves[1]).astype(float),
    const=1.0)
farm_pooled = clustered_ols(farm_pooled_X, farm_log['
    partial_value'])

```

5.3.11 Two waves: remove persistent omitted factors

Suppose the composite omitted term is $u_{it} = \alpha_i + \delta_t + \epsilon_{it}$. Then

$$\Delta y_i = (\delta_2 - \delta_1) + \beta_H \Delta h_i + \beta_L \Delta \ell_i + \Delta \epsilon_i.$$

- Persistent management, land quality, and price differences cancel *if their contributions are additive and constant*.
- The intercept captures the common time change.
- Shocks that change input choices, changing prices, and measurement error remain potential sources of bias.

We need $E[\Delta \epsilon_i \mid z_{i1}, z_{i2}, \alpha_i] = 0$ in the estimation sample, plus identifying variation.

5.3.12 Estimate changes and compare

With two complete waves, first differences with an intercept give the TWFE slopes. Household demeaning alone leaves the time effect behind. Subtract by household labels, checking that both waves contain the same households in the same order.

```
farm_first = farm_log.xs(farm_waves[0], level='t').
    sort_index()
farm_second = farm_log.xs(farm_waves[1], level='t').
    sort_index()
assert farm_first.index.equals(farm_second.index)
farm_difference = farm_second - farm_first
farm_fd_X = farm_difference[farm_xcols].assign(const=1.0)
farm_fd = clustered_ols(farm_fd_X, farm_difference['
    partial_value'])
farm_models = {**farm_cross, 'pooled_+wave': farm_pooled
    , 'first_differences': farm_fd}
farm_comparison = pd.DataFrame({
    name: pd.Series({term: b[term] for term in farm_xcols
    })
    for name, (b, V) in farm_models.items()})
farm_se = pd.DataFrame({
    name: pd.Series({term: np.sqrt(V.loc[term, term]) for
    term in farm_xcols})
    for name, (b, V) in farm_models.items()})
print("Slopes;_same", len(farm_ids), "farms_in_every_
    model")
display(farm_comparison.round(3))
print("Farm-cluster_CR1_standard_errors")
display(farm_se.round(3))
```

A difference between the columns isn't an estimate of management ability. It also reflects the different variation used, measurement error, and possible departures from the common-slope model. Restricting to a common sample removes one source of difference, not all of them.

Repeat the differenced fit for the same-crop-set subsample. Report its size and compare coefficients; don't choose the sample because its coefficient looks more plausible.

```
farm_stable = farm_difference.loc[farm_stable_ids].
    sort_index()
farm_stable_X = farm_stable[farm_xcols].assign(const=1.0)
if len(farm_stable) > len(farm_stable_X.columns) and (
    np.linalg.matrix_rank(farm_stable_X) == len(
    farm_stable_X.columns)):
```

```

farm_stable_fit = clustered_ols(farm_stable_X,
    farm_stable['partial_value'])
display(pd.DataFrame({'all_farms': farm_fd[0],
    'same_crop_set':
        farm_stable_fit[0]}).round
    (3))
print("Same-crop-set_farms:", len(farm_stable))
else:
    print("Too_few_independent_input_changes_for_the_same
        -crop-set_fit")

```

5.3.13 Farm means: a related specification check

Mundlak's later (1978) formulation augments the levels regression with household means of the regressors. On this balanced, unweighted panel, including those means *and the wave indicator* reproduces the TWFE slopes. This is an algebra check on the implementation.

Test the two mean coefficients jointly using farm-cluster covariance. Rejection is evidence against the corresponding pooled-model restriction, not proof that management caused the discrepancy. Non-rejection doesn't establish exogeneity or show that the panel was unnecessary. The reported chi-squared p-value is an asymptotic approximation over independent farms; it isn't an exact test based on the two survey dates.

```

farm_means = farm_log[farm_xcols].groupby('i').transform(
    'mean').add_suffix('_mean')
farm_cre_X = pd.concat([farm_pooled_X, farm_means], axis
    =1)
farm_cre_b, farm_cre_V = clustered_ols(farm_cre_X,
    farm_log['partial_value'])
np.testing.assert_allclose(farm_cre_b[farm_xcols],
    farm_fd[0][farm_xcols],
        rtol=1e-9, atol=1e-9)
farm_mean_terms = farm_means.columns
farm_gamma = farm_cre_b[farm_mean_terms].to_numpy()
farm_gamma_V = farm_cre_V.loc[farm_mean_terms,
    farm_mean_terms].to_numpy()
assert np.linalg.matrix_rank(farm_gamma_V) == len(
    farm_mean_terms)
farm_wald = float(farm_gamma @ np.linalg.solve(
    farm_gamma_V, farm_gamma))
display(pd.DataFrame({'coefficient': farm_cre_b,
    'cluster_se': np.sqrt(np.diag(
        farm_cre_V))}).round(3))

```

```
print("Joint_mean-term_Wald_chi-square:", round(farm_wald
, 3),
      ";_df:", len(farm_mean_terms),
      ";_p-value:", chi2.sf(farm_wald, len(
farm_mean_terms)))
```

5.3.14 What changes when we follow farms?

Same 506 farms; nominal partial crop value; unweighted estimates:

Log input	Pooled + wave	First differences
Crop area	0.660	0.363
Labor	0.297	0.091

- Both slopes fall; the farm-mean restriction is rejected.
- Partial output and unstable area remain problems.

This doesn't isolate a causal management effect.

5.3.15 Read the estimates with the diagnostics

On the checked extracts, the land coefficient is 0.507 in the first cross-section and 0.886 in the second. Pooling with a wave indicator gives 0.660; first differences give 0.363. The corresponding pooled and differenced labor slopes are 0.297 and 0.091, with farm-cluster standard errors of 0.052 and 0.091. These describe the same 506 farms throughout.

The joint farm-mean test has chi-square 49.335 on two degrees of freedom (p approximately 1.94×10^{-11}), under independent-farm asymptotics. This rejects the tested pooled restriction; it doesn't establish the differenced model's exogeneity condition. With only 21 same-crop-set farms, the differenced land slope is 0.631 and the labor slope is -0.142. That change is another reason to investigate the area measure, not a reason to select this small sample as the preferred answer.

Time-constant additive effects have been removed. Missing own-consumed output, heterogeneous prices, changes in management, and mismeasured inputs have not. The two-period panel gives us a different comparison, whose assumptions must still be defended.

5.3.16 FWL in an unbalanced panel

Stack the observed rows and write $y = Z\beta + D\theta + \epsilon$, where D contains farm and wave indicators. Define

$$M_D = I - D(D'D)^+D'.$$

- $M_D D = 0$: both sets of fixed effects disappear.
- Regress $M_D y$ on $M_D Z$ for the same slopes as full OLS.
- Use the *same observed rows* in every projection.
- One pass subtracting farm and wave means generally fails when the panel is unbalanced.

There is no third GLSS panel wave. Our example is simulated.

5.3.17 Check FWL on an unbalanced example

This illustration has eight farms and four dates, with a fixed pattern of missing observations. It isn't additional Ghana data. We use `DataMat.resid` to remove farm and wave indicators jointly, then check that residual regression and full indicator OLS agree. A seeded outcome contains noise, so the check is agreement between estimators, not exact recovery of the generating coefficient.

```
farm_rng = np.random.default_rng(20260916)
fwl_index = pd.MultiIndex.from_product([range(8), range(4)], names=['i', 't'])
fwl_index = fwl_index[np.arange(len(fwl_index)) % 7 != 0]
fwl_Z = pd.DataFrame(farm_rng.normal(size=(len(fwl_index), 2)),
                     index=fwl_index, columns=['x1', 'x2'])
fwl_D = pd.get_dummies(fwl_index.to_frame(index=False).astype(str),
                       drop_first=True, dtype=float)
fwl_D.index = fwl_index
fwl_D = fwl_D.assign(const=1.0)
fwl_y = pd.Series(fwl_Z.to_numpy() @ np.array([.4, .7])
                  + fwl_D.to_numpy() @ farm_rng.normal(
                      size=fwl_D.shape[1])
                  + farm_rng.normal(scale=.1, size=len(fwl_index)),
```

```

        index=fwl_index, name='y')
fwl_joint = dm.DataMat(fwl_Z.assign(y=fwl_y))
fwl_residual = pd.DataFrame(fwl_joint.resid(dm.DataMat(
    fwl_D)))
fwl_partial = dm.DataMat(fwl_residual.iloc[:, :2]).lstsq(
    dm.DataVec(fwl_residual.iloc[:, 2]))
fwl_full = dm.DataMat(pd.concat([fwl_Z, fwl_D], axis=1)).
    lstsq(dm.DataVec(fwl_y))
np.testing.assert_allclose(np.asarray(fwl_partial), np.
    asarray(fwl_full)[:2],
                                atol=1e-10, rtol=1e-10)
np.testing.assert_allclose(fwl_D.to_numpy().T @
    fwl_residual.to_numpy(),
                                0, atol=1e-10)
fwl_naive = (fwl_Z - fwl_Z.groupby('i').transform('mean')
    - fwl_Z.groupby('t').transform('mean') +
    fwl_Z.mean())
print("Rows_per_farm:", fwl_Z.groupby('i').size().to_dict
    ())
print("Full_OLS_slopes:", np.asarray(fwl_full)[:2])
print("FWL_slopes:", np.asarray(fwl_partial))
print("One-pass_means_equal_joint_residuals:",
    np.allclose(fwl_naive, fwl_residual.iloc[:, :2]))

```

The explicit indicator matrix is small enough here to make the algebra visible. Large panels need methods that absorb effects without forming a dense observation-by-observation projection matrix. Slope equivalence doesn't automatically give the correct covariance or degrees of freedom.

5.4 Lifecycle consumption

5.4.1 Why study consumption over a lifetime?

Income and needs change over a lifetime. Consumption needn't move one-for-one with current income.

- Saving transfers resources from today to later ages.
- Borrowing transfers future resources toward today, when credit is available.
- Uncertain earnings, health, and longevity affect those choices.

The life-cycle model asks how households allocate resources across ages. It motivates an age profile of consumption; it doesn't determine that profile without assumptions about preferences and constraints.

5.4.2 The life-cycle budget

At age a , choose consumption c_a and assets to carry forward:

$$b_{a+1} = (1 + r)b_a + y_a - c_a.$$

Here b_a is assets, y_a income, and r the interest rate.

- Choices trade current against expected future utility, given initial assets and a requirement to repay debts by the end of life.
- Credit limits can prevent the desired transfer of resources.
- Uninsured shocks can make households' consumption paths diverge.

We need data across ages to study these implications.

5.4.3 An application: inequality as cohorts age

Deaton and Paxson (1994) follow consumption inequality among people born around the same time. Our question is narrower:

Does dispersion in food purchases per household member rise as a cohort ages?

- The mean describes the cohort's average log purchases.
- The variance describes differences *among* its households.
- We compare the same birth cohorts across survey waves.

Food purchases omit own-produced food and non-food consumption. This exercise doesn't measure total consumption inequality.

5.5 Panels from Cross-Sections

5.5.1 Following groups instead of households

Most GLSS rounds interview a new sample of households. We can't compute an individual household's change between those rounds.

We can follow a *birth cohort*: for example, households whose heads were born in 1960–64.

- Find representatives of that cohort in each survey.
- Summarize their circumstances using the survey weights.
- Follow those summaries as the cohort ages.

The group definition stays fixed; the sampled households change.

5.5.2 A pseudo-panel

For cohort c in wave t , estimate the population mean with

$$\bar{y}_{ct} = \frac{\sum_{i \in (c,t)} w_{it} y_{it}}{\sum_{i \in (c,t)} w_{it}}.$$

- One row now describes a cohort and a date.
- Repeated rows let us study changes in cohort means or dispersion.
- Small cells give noisy summaries, even when the whole survey is large.

This is a panel of *estimated group characteristics*, not a household panel. It can't reveal which particular households moved up or down.

5.5.3 From cells to a regression

A working model for cohort means is

$$\bar{y}_{ct} = \alpha_c + \delta_t + \beta' \bar{z}_{ct} + u_{ct}.$$

Age terms and household composition can enter $\bar{z}_{ct}$.

- α_c captures persistent differences between cohorts.
- δ_t captures changes common to cohorts at a survey date.
- Both outcome and regressor means contain sampling error.

Larger cells help, but don't establish that the regression error is unrelated to the regressors. We'll first inspect the cohort profiles.

5.5.4 Identification: age, cohort, and time

For a birth year b , age a , and calendar year t ,

$$a = t - b.$$

The data cannot assign a linear trend uniquely to age, cohort, or time.

- Following a cohort changes both its age and the calendar year.
- Comparing ages in one survey also compares birth cohorts.
- Separating all three requires an explicit restriction.

Cohort paths aren't pure age effects.

5.5.5 What makes cohorts comparable?

A birth year doesn't change as a person ages. Headship can: households form, split, and change heads. Migration and mortality can also change who represents a cohort in successive surveys. A fixed label therefore doesn't guarantee a fixed population of households.

The sampling error in a cohort mean gets smaller as the cell grows, under suitable sampling conditions. If both the outcome and regressors are estimated cell means, their sampling errors may be correlated. Measurement error then complicates regression; it needn't produce simple attenuation of every coefficient. We begin with descriptive profiles.

5.5.6 Start with household observations

We need food expenditure, household size, the head's age, and survey weights. These are the same survey tables used earlier. Ask for food expenditure in constant-2017 local currency so inflation and the change in Ghana's currency units aren't mistaken for changes in purchasing.

```
import warnings
warnings.filterwarnings("ignore", message=r"(?s).*Set_
    LSMS_[A-Z_]+=1_to_make_this_fatal")
warnings.filterwarnings("ignore", message=r"
    household_characteristics: replaced.*")
warnings.filterwarnings("ignore", message=r"Refusing_
    workspace_copy_of_DVC-tracked_file.*")

import lsms_library as ll
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from IPython.display import display

ghana = ll.Country('GhanaLSS')
roster = ghana.household_roster()
chars = ghana.household_characteristics()
sample = ghana.sample()
food = ghana.food_expenditures(numeraire='LCU-real-2017')
```

First make one observation per household and wave. Sum expenditure over food items. The characteristics table already provides log household size, so subtracting it from log expenditure gives log expenditure per member. This is a per-member comparison, not an adjustment for different needs of adults and children.

For age, keep households with exactly one recorded head. A missing or ambiguous head cannot define our cohort. The survey weight belongs to the household, so the averages below describe *households* classified by the head's birth year; they aren't person-weighted averages.

```
heads = roster[roster['Relationship'].str.strip().str.
    lower().eq('head')]
head_count = heads.groupby(['t', 'i']).size()
age = heads.groupby(['t', 'i'])['Age'].first().rename('
    age')
age = age[head_count.eq(1)]
expenditure = food['Expenditure'].groupby(['t', 'i']).sum
    (min_count=1).rename('food')
log_size = chars['log_HSize'].groupby(['t', 'i']).first()
    .rename('log_size')
weights = sample['weight'].reorder_levels(['t', 'i']).
    sort_index().rename('weight')

cohort_data = pd.concat([expenditure, log_size, age,
    weights], axis=1)
cohort_data = cohort_data.replace([np.inf, -np.inf], np.
    nan).dropna()
cohort_data = cohort_data[cohort_data['food'].gt(0)
    & cohort_data['weight'].gt(0)
    & cohort_data['age'].between
    (25, 70)].copy()
cohort_data['log_food_pc'] = (np.log(cohort_data['food'].
    astype(float))
    - cohort_data['log_size'])
display(cohort_data.groupby('t').agg(households=('age', '
    size'),
    mean_age=('age', '
    mean')).round(1))
```

We use the final year named in each survey wave as its reference year. Subtract the head's reported age to approximate the birth year. The surveys span more than one calendar year and ages can be rounded, so a head close to a cohort boundary can be misclassified.

Five-year cohorts group estimated birth years 1960–64 together, 1965–69 together, and so on. Dividing by five and taking the floor finds the group; multiplying by five gives its first birth year.

```
wave_year = {wave: int(wave[:4]) + 1 for wave in ghana.
    waves}
cohort_data['year'] = cohort_data.index.get_level_values(
```

```

    't').map(wave_year)
cohort_data['born'] = cohort_data['year'] - cohort_data['age']
cohort_data['cohort'] = (cohort_data['born'] // 5 * 5).
    astype(int)
display(cohort_data[['age', 'year', 'born', 'cohort']].
    head())

```

5.5.7 Calculate cohort means and dispersion

Let ℓ_i be log real food expenditure per member. Within each cohort-wave cell, calculate

$$\bar{\ell} = \frac{\sum_i w_i \ell_i}{\sum_i w_i}, \quad v = \frac{\sum_i w_i (\ell_i - \bar{\ell})^2}{\sum_i w_i}.$$

The second quantity is a weighted descriptive variance among households. It isn't the variance of the estimated mean or a regression standard error. Notice also that we average *logs*, rather than take the log of average expenditure.

Put these steps in a function so we can change the cohort width without changing how we calculate the summaries. The final two conditions omit cells whose birth band is only partly covered by our age range, 25–70.

```

def cohort_summary(data, width=5):
    d = data.reset_index().copy()
    d['cohort'] = (d['born'] // width * width).astype(int)
    d['weighted_log'] = d['weight'] * d['log_food_pc']
    d['weighted_age'] = d['weight'] * d['age']
    groups = d.groupby(['cohort', 't'])
    total_weight = groups['weight'].sum()
    mean_for_row = (groups['weighted_log'].transform('sum')
        / groups['weight'].transform('sum'))
    d['weighted_deviation2'] = d['weight'] * (d['log_food_pc'] - mean_for_row)**2
    groups = d.groupby(['cohort', 't'])
    result = pd.DataFrame({
        'mean_log': groups['weighted_log'].sum() /
            total_weight,
        'var_log': groups['weighted_deviation2'].sum() /
            total_weight,
        'age': groups['weighted_age'].sum() /
            total_weight,
    })

```

```

        'n': groups.size(),
        'year': groups['year'].first(),
    })
    birth_start = result.index.get_level_values('cohort')
    youngest = result['year'] - (birth_start + width - 1)
    oldest = result['year'] - birth_start
    return result[youngest.ge(25) & oldest.le(70)]

cells = cohort_summary(cohort_data, width=5)
display(cells.head(12).round(3))

```

5.5.8 Check the cell sizes

Before drawing a line through cohort summaries, inspect the observations behind each point. We'll display cells with at least 100 households. This is a choice for the exercise, not a guarantee of precision: unequal weights and sampling clusters also affect precision.

A cohort needs at least three displayed waves to give us more than one change to look at. A blank cell in the count table means that the cohort isn't fully covered by our age window in that wave.

```

minimum_cell = 100
display(cells['n'].unstack('t').astype('Int64'))
print("Cells meeting the display cutoff:", int(cells['n'
    ].ge(minimum_cell).sum()),
      "of", len(cells))
big = cells[cells['n'].ge(minimum_cell)].reset_index()
number_of_waves = big.groupby('cohort')['t'].nunique()
shown_cohorts = number_of_waves.index[number_of_waves.ge
    (3)]
big = big[big['cohort'].isin(shown_cohorts)]

```

5.5.9 Follow the same cohorts across surveys

Each line below represents a fixed birth-year band. The left panel shows average log real food purchases per member; the right shows the variance of those logs. Read the axes before comparing the shapes: a cohort can have rising average purchases while differences among its households also grow.

```

fig, axes = plt.subplots(1, 2, figsize=(11, 5), sharex=
    True)
for ax in axes:
    ax.set_prop_cycle(color=plt.cm.tab20.colors)
for cohort, group in big.groupby('cohort'):

```

```

group = group.sort_values('year')
label = f"{cohort}-{cohort+4}"
axes[0].plot(group['age'], group['mean_log'], 'o-',
             label=label)
axes[1].plot(group['age'], group['var_log'], 'o-',
             label=label)
axes[0].set_ylabel('Mean_log_purchases_per_member')
axes[1].set_ylabel('Variance_of_log_purchases_per_member')
)
for ax in axes:
    ax.set_xlabel('Mean_age_of_household_head')
    ax.grid(alpha=.2)
fig.legend(*axes[1].get_legend_handles_labels(), title='
Birth_cohort',
          loc='lower_center', ncol=6, fontsize=8,
          frameon=False)
fig.tight_layout(rect=[0, .2, 1, 1])
plt.show()

```

Where a line rises in the right panel, within-cohort dispersion rises between those surveys. That interval also contains changes in prices, policy, measurement, and who heads a household. A vertical difference between lines isn't an identified cohort effect, and overlapping lines don't establish that cohort effects are absent.

This is a descriptive application inspired by Deaton and Paxson. Food purchases are narrower than consumption: changing reliance on own-grown food can change the graph without the same change in living standards. Dividing by household size also leaves differences in household needs.

5.5.10 Compare changes and cohort widths

Summarize the change from the first to the last displayed wave for each cohort. Keep the years and ages beside the change: not every cohort is observed over the same interval. Dividing by years gives a descriptive annual change, not an estimated causal age effect.

```

ordered = big.sort_values(['cohort', 'year'])
first = ordered.groupby('cohort').first()
last = ordered.groupby('cohort').last()
cohort_changes = pd.DataFrame({
    'first_year': first['year'], 'last_year': last['year'],
    },
    ['first_age': first['age'], 'last_age': last['age'],

```

```

        'variance_change': last['var_log'] - first['var_log',
    ],
    })
    cohort_changes['change_per_year'] = (cohort_changes['
        variance_change']
        / (cohort_changes['last_year'] - cohort_changes['
            first_year']))
    display(cohort_changes.round(3))

```

Here all eleven displayed five-year cohorts have lower dispersion at their last displayed date than at their first. Several paths rise and fall in between. These food-purchase profiles don't show a general increase in dispersion as cohorts age, and they don't by themselves test a model of total consumption or identify an age effect.

Now widen the birth-year bands to ten years. Larger cells usually reduce sampling noise, but combine people with more different ages. Repeat the same calculation and cutoff; don't choose the width because one graph looks closer to a prediction.

```

cells10 = cohort_summary(cohort_data, width=10)
fig, axes = plt.subplots(1, 2, figsize=(11, 5), sharey=
    True)
for ax, width, table in zip(axes, [5, 10], [cells,
    cells10]):
    ax.set_prop_cycle(color=plt.cm.tab20.colors)
    visible = table[table['n'].ge(minimum_cell)].
        reset_index()
    for cohort, group in visible.groupby('cohort'):
        if group['t'].nunique() >= 3:
            group = group.sort_values('year')
            ax.plot(group['age'], group['var_log'], 'o-',
                label=f"{cohort}-{cohort+width-1}
                    ")
    ax.set_title(f'{width}-year_birth_cohorts')
    ax.set_xlabel('Mean_age_of_household_head')
    ax.legend(fontsize=8, frameon=False, ncol=3,
        loc='upper_center', bbox_to_anchor=(.5,
            -.2))
    ax.grid(alpha=.2)
axes[0].set_ylabel('Variance_of_log_purchases_per_member'
    )
fig.tight_layout()
plt.show()
display(pd.DataFrame({
    '5-year': cells['n'].describe(), '10-year': cells10['

```

```

        n'].describe()
    }).round(1))

```

Read the two graphs together. Do they support a general rise in within-cohort food-purchase dispersion, or do some paths fall? Which changes coincide across cohorts at a survey date? The exercise provides these comparisons; separating a life-cycle mechanism from period changes would require additional assumptions and a broader consumption measure.

5.6 Household panels and welfare

5.6.1 The Uganda panel

```

uga = ll.Country('Uganda')
print(uga.waves)
print(uga.data_scheme)

ufood = uga.food_expenditures()
uc = ufood.groupby(['t', 'i']).sum().squeeze()
uc = np.log(uc.where(uc > 0)).rename('logc')

un = uga.household_characteristics().sum(axis=1).groupby(['t', 'i']).first()
un = np.log(un.astype(float).where(un > 0)).rename('logn')

p = pd.concat([uc, un], axis=1).dropna()
p = p[np.isfinite(p).all(axis=1)]

# How many households are actually observed more than once?
counts = p.groupby('i').size()
print(counts.value_counts().sort_index())

# Keep only households seen more than once — singletons contribute nothing
# to a within estimator, and quietly inflate the apparent sample size.
p = p[counts.reindex(p.index.get_level_values('i')).to_numpy() > 1]

# Time dummies, then demean everything within household.
After demeaning
# there is no constant to include.
T = pd.get_dummies(p.index.get_level_values('t'), prefix='t',
                    drop_first=True, dtype=float)
T.index = p.index

```

```

q = pd.concat([p, T], axis=1)
qd = q - q.groupby(level='i').transform('mean')

# The farm application's estimator, reused: least squares with a CR1
# covariance clustered on the index level 'i' — here the household.
b, V = clustered_ols(qd.drop(columns='logc'), qd.logc)
print(coef_table(b, V).to_string())

```

Compare the coefficient on `logn` here with the cross-sectional one from session 4. If they differ, the cross-sectional estimate was contaminated by whatever it is about large households that also predicts consumption.

5.6.2 The same panel, in welfare units

Everything above treats `logc` as the outcome. Session 4 argued that $w = -\log \lambda$ is the better welfare measure, because prices and household composition are swept into the good-time effects and the Barten scales rather than left in the number. Here is the same regression on w , so you can see how much it matters.

```

from pathlib import Path
import lsms_library as ll
import cfe
from cfe import Regression
import numpy as np, pandas as pd

def uganda_cfe():
    # The estimated CFE system for Uganda, as a cfe.Regression.
    The object
    # carries beta, gamma, w = -log lambda, and predicted expenditures, so
    # nothing downstream has to refit. Uses a copy staged on the hub if
    # there is one, else a copy you estimated earlier, else estimates it
    # from scratch (about forty seconds) and caches the result.
    So this
    # cell is self-contained: no other notebook need have been run first.
    staged = Path('/srv/data/hhsurveys/uganda.rgsn')
    mine    = Path.home() / '.cache' / 'hhsurveys' / 'uganda.rgsn'
    for c in (staged, mine):
        if c.exists():
            return cfe.read_pickle(str(c))

```

```

uga = ll.Country('Uganda')
x = uga.food_expenditures().squeeze()
agg = (uga.categorical_mapping['harmonize_food']
        .set_index('Preferred_Label')['Aggregate_Label'].to_dict())
x = x.rename(index=agg, level='j')
x = x.groupby(x.index.names).sum()

y = np.log(x.replace(0, np.nan).dropna()).groupby(['i', 't', 'j']).sum()
y = pd.concat({1: y}, names=['m']).reorder_levels(['i', 't', 'm', 'j'])

d0 = uga.household_characteristics()
d = d0.assign(
    Girls=d0[[f'F_{a}' for a in ['00-03', '04-08', '09-13', '14-18']]],
    Boys =d0[[f'M_{a}' for a in ['00-03', '04-08', '09-13', '14-18']]],
    Women=d0[[f'F_{a}' for a in ['19-30', '31-50', '51+']]].sum(axis=1),
    Men  =d0[[f'M_{a}' for a in ['19-30', '31-50', '51+']]].sum(axis=1)
)[['Girls', 'Boys', 'Women', 'Men', 'log_HSize']].dropna(how='any')
d = d.groupby(['i', 't']).first()
d = pd.concat({1: d}, names=['m']).reorder_levels(['i', 't', 'm']).sort

r = Regression(y=y, d=d)
r.get_beta(); r.get_w(); r.predicted_expenditures()
mine.parent.mkdir(parents=True, exist_ok=True)
r.to_pickle(str(mine))
return r

r = uganda_cfe()
w = r.get_w()
w.groupby('t').agg(['size', 'mean', 'std']).round(3)

# The same within estimator, in welfare units.
# w arrives indexed (i, t, m) while p is (t, i). Line them up explicitly:
# concat on a differently-ordered MultiIndex does not raise, it just fails
# to align — the same trap that emptied the pseudo-panel above.
wi = w.droplevel('m') if 'm' in w.index.names or [] else w
wi = wi.rename('w').reorder_levels(['t', 'i']).sort_index()

q2 = pd.concat([wi, p.logn], axis=1).dropna()
assert len(q2) > 0.5 * min(len(wi), len(p)), "w_and_p_failed_to_align"

```

```

c2 = q2.groupby('i').size()
q2 = q2[q2.reindex(q2.index.get_level_values('i')).to_numpy() > 1]

T2 = pd.get_dummies(q2.index.get_level_values('t'), prefix='t',
                    drop_first=True, dtype=float)
T2.index = q2.index
Q2 = pd.concat([q2, T2], axis=1)
Q2d = Q2 - Q2.groupby(level='i').transform('mean')

b_w, V_w = clustered_ols(Q2d.drop(columns='w'), Q2d.w)
print(coef_table(b_w, V_w).to_string())

```

Three comparisons to make, and the third is the one to remember.

The coefficient on **logn**. In $\log c$ it is about +0.34: a bigger household simply spends more. In w it is about -0.03 — small and *negative*. Household size has already been absorbed into the Barten scales, so what is left is what size does to welfare net of need, and that is slightly adverse. The two numbers are not rival estimates of one parameter; they answer different questions.

The time effects. In $\log c$ they climb steeply and monotonically, from 0.40 to 1.20 across the panel. That is mostly inflation, since nothing deflated those expenditures. In w they sit within ± 0.15 and do not trend, because prices live in a_t^j and never entered w at all. No CPI was used to achieve that, and none was available.

Now look at 2009–10 in the w column. It is the only negative time effect in the panel. That is the 2008 food price crisis, and it is invisible in $\log c$, where 2009–10 is simply another step up the inflation ladder. Session 6 takes that observation and asks what it does to the poverty rate.

5.6.3 Exercises

1. Rebuild the Ghana pseudo-panel with ten-year cohorts. How much do the lifecycle profiles change? Which conclusion is robust?
2. Compare the weighted mean of log purchases with the log of weighted mean purchases in each cohort. Why are they different?
3. In Uganda, quantify attrition: what fraction of the 2009–10 households appear in 2011–12? Are the leavers different in 2009–10 consumption from the stayers? What does that do to your fixed-effects estimate?

6 Capstone and Frontiers

Where the methods break down, and then your own analysis.

6.1 Reading

6.1.1 Reading

- Ligon, "Household Welfare from Disaggregate Demands" (working paper)
- Ligon (2026), "Risk sharing tests and covariate shocks: Drought, Floods, and Pests in Uganda," *AER*, forthcoming
- Townsend (1994), "Risk and insurance in village India," *Econometrica*

6.2 Two measures of poverty, and they disagree

Take Uganda through the 2008 food price crisis, when cereal prices more than doubled. Anchor a headcount at 58% in 2005–06 and carry the *same* threshold forward.

- on $w = -\log \lambda$: poverty **rises** to about 63% by 2009–10
- on nominal per-capita expenditure: the median rises about 32% over the same interval

Deflate by anything less than 32% inflation and you conclude that welfare improved. The two measures disagree in **sign**, not in magnitude.

6.3 Two measures of poverty, and they disagree

This is the reason session 4 spent time on rank, and it is worth stating the mechanism rather than leaving it as a surprising number.

Deflating nominal expenditure by a price index answers the question "how much more of the 2005 bundle could this household buy?" When relative prices move — cereals doubling while other prices do not — that question stops tracking welfare, because the household does not want the 2005 bundle any more, and because the bundle it does want has become more expensive in a way no single scalar records. With rank at least three, three indices would be needed; with one you get an answer whose error is correlated with exactly the price movement you are trying to measure.

The λ estimate never forms an index. It reads welfare off the composition of expenditure directly, and the composition is precisely what responds to relative prices. That is why it can move in the opposite direction, and why, when the two disagree, the burden of proof is not on it.

6.4 Full risk sharing

If a group of households shares risk fully, they equate marginal utilities of expenditure up to fixed Pareto weights:

$$\lambda_{it} = \frac{\mu_t}{\theta_i} \iff w_{it} = \log \theta_i - \log \mu_t.$$

So under full insurance w_{it} is a household effect plus a time effect, and **nothing else**. The test writes itself:

$$w_{it} = \alpha_i + \delta_t + \gamma' S_{it} + \varepsilon_{it},$$

with S_{it} household-level shocks. Full risk sharing implies $\gamma = 0$.

6.5 Why covariate shocks break the test

The test above has a hole, and it is exactly where the policy interest is.

- δ_t absorbs **everything** common to the group in period t
- a drought that hits the whole region *is* common to the group
- so the covariate component of the shock is differenced away by the very time effect that makes the test valid

6.6 What the risk-sharing test can establish

The standard test has power only against **idiosyncratic** shocks.

Failing to reject tells you nothing about insurance against common drought, flood, or pest shocks. These are the risks a village cannot self-insure by transfers among its members.

6.7 Why covariate shocks break the test

There is a general lesson here that outlives this application. A fixed effect is not free: it removes variation, and you have to ask what was in the variation it removed. Session 5 made the point in passing — household effects absorb

time-invariant selection and nothing else — but here the cost is sharper, because the absorbed variation is not a nuisance. It is the treatment.

The way out is not econometric cleverness so much as data: identify who was exposed to the covariate shock and who was not, using something outside the survey's own time dimension — rainfall, remote-sensed vegetation, the spatial footprint of a flood — and then the exposure varies within a period and survives the time effect. That is what the paper does, and it is the reason it needs the LSMS-ISA plot data alongside the panel.

6.8 The sign flip, computed

```
# Show tracebacks without the library's internal frames: the line that
# failed, and why. Change Plain to Verbose if you ever want the rest.
%xmode Plain

# The library audits its own corpus on first read and reports what it finds
# — implausible quantities, NaN index keys, a column that is wholly null
# one wave — at multi-paragraph length. Those reports are a work queue for
# whoever maintains the data, not something the room can act on, and they block
# the output they are attached to. Silenced here by their own
# "Set LSMS_..._STRICT=1" signature, which is precise: every other warning,
# pandas deprecations included, still shows. Delete these two lines to read
# them.
import warnings
warnings.filterwarnings("ignore",
                        message=r"(?s).*Set_LSMS_[A-Z_]+=1_to_make_this_fatal")

# The first call that builds a table may print "DVC unavailable ...
# falling back to manual aggregation". That is about how the library
# fetches its raw files, not about your data; the numbers are the same,
# and it does not recur once the table is built.

import matplotlib as mpl
mpl.rcParams.update({'figure.dpi': 110, 'font.size': 11,
                    'lines.linewidth': 1.8})

from pathlib import Path
import lsms_library as ll
import cfe
from cfe import Regression
```

```

import numpy as np, pandas as pd

def uganda_cfe():
    # The estimated CFE system for Uganda, as a cfe.Regression.
    The object
    # carries beta, gamma, w = -log lambda, and predicted expenditures, so
    # nothing downstream has to refit. Uses a copy staged on the hub if
    # there is one, else a copy you estimated earlier, else estimates it
    # from scratch (about forty seconds) and caches the result.
    So this
    # cell is self-contained: no other notebook need have been run first.
    staged = Path('/srv/data/hhsurveys/uganda.rgsn')
    mine = Path.home() / '.cache' / 'hhsurveys' / 'uganda.rgsn'
    for c in (staged, mine):
        if c.exists():
            return cfe.read_pickle(str(c))

    uga = ll.Country('Uganda')
    x = uga.food_expenditures().squeeze()
    agg = (uga.categorical_mapping['harmonize_food']
           .set_index('Preferred_Label')['Aggregate_Label'].to_dict())
    x = x.rename(index=agg, level='j')
    x = x.groupby(x.index.names).sum()

    y = np.log(x.replace(0, np.nan).dropna()).groupby(['i', 't', 'j']).sum()
    y = pd.concat({1: y}, names=['m']).reorder_levels(['i', 't', 'm', 'j'])

    d0 = uga.household_characteristics()
    d = d0.assign(
        Girls=d0[['F_{a}'] for a in ['00-03', '04-08', '09-13', '14-18']],
        Boys=d0[['M_{a}'] for a in ['00-03', '04-08', '09-13', '14-18']],
        Women=d0[['F_{a}'] for a in ['19-30', '31-50', '51+']].sum(axis=1),
        Men=d0[['M_{a}'] for a in ['19-30', '31-50', '51+']].sum(axis=1)
    )[['Girls', 'Boys', 'Women', 'Men', 'log_HSize']].dropna(how='any')
    d = d.groupby(['i', 't']).first()
    d = pd.concat({1: d}, names=['m']).reorder_levels(['i', 't', 'm']).sort

    r = Regression(y=y, d=d)
    r.get_beta(); r.get_w(); r.predicted_expenditures()
    mine.parent.mkdir(parents=True, exist_ok=True)

```

```

        r.to_pickle(str(mine))
    return r

r = uganda_cfe()
w = r.get_w()

waves = sorted(w.index.get_level_values('t').unique())
base = waves[0]

# Anchor at 58% in the base wave, then carry the SAME welfare threshold
# forward. This is the whole exercise: one line, one threshold, no index.
z = w.xs(base, level='t').quantile(0.58)
hc = w.groupby('t').apply(lambda s: float((s <= z).mean()))
print(f"headcount_on_w, anchored_at_58%_in_{base}:")
print(hc.round(3).to_string())

# The conventional measure, undeflated: nominal per-capita expenditure.
uga = ll.Country('Uganda')
xt = uga.food_expenditures().squeeze().groupby(['i', 't']).sum()
n = np.exp(uga.household_characteristics()['log_HSize']).groupby(['i', 't'])
pc = (xt / n.reindex(xt.index)).dropna()

med = pc.groupby('t').median()
print("median_nominal_per-capita_food_expenditure:")
print(med.round(0).to_string())
print(f"\nchange_{waves[0]}->_{waves[1]}:_"
      f"{100*_((med.loc[waves[1]]/_med.loc[waves[0]]-_1):+.0f}%")
print(f"headcount_on_w, same_interval:_"
      f"{hc.loc[waves[0]]:.3f}>_{hc.loc[waves[1]]:.3f}")

```

One measure says welfare fell; the other says spending rose by a third. Reconciling them requires a deflator larger than the measured inflation over the period, which is to say: the conventional answer depends entirely on a number the survey does not contain.

6.9 A risk-sharing test

```
import datamat as dm
```

```
# Session 5's estimator, repeated so this notebook stands on its own:
```

```

# least squares with a CR1 covariance clustered on the index level 'i'.
def clustered_ols(X, y):
    assert X.index.equals(y.index)
    X, y = X.astype(float), y.astype(float)
    n, k = X.shape
    assert np.isfinite(X.to_numpy()).all() and np.isfinite(y).all()
    assert np.linalg.matrix_rank(X.to_numpy()) == k and n > k
    b = pd.Series(np.asarray(dm.DataMat(X).lstsq(dm.DataVec(y))).reshape(-1,
                                index=X.columns)
    residual = y - X @ b
    scores = X.mul(residual, axis=0).groupby(level='i').sum()
    G = len(scores)
    assert G > 1
    bread = np.linalg.inv(X.to_numpy().T @ X.to_numpy())
    V = bread @ (scores.to_numpy().T @ scores.to_numpy()) @ bread
    V *= (G / (G - 1)) * ((n - 1) / (n - k))
    return b, pd.DataFrame(V, index=X.columns, columns=X.columns)

def coef_table(b, V):
    """Coefficients, cluster-robust standard errors, and t statistics."""
    se = pd.Series(np.sqrt(np.diag(V)), index=b.index)
    return pd.DataFrame({'coef': b, 'std_err': se, 't': b / se}).round(3)

sh = uga.shocks()
print(sh.index.get_level_values('Shock').unique().tolist())

# One indicator per shock type, per household-wave.
S = (sh.assign(hit=1.0).hit
      .groupby(['i', 't', 'Shock']).max()
      .unstack('Shock').fillna(0.0))

wi = w.droplevel('m') if 'm' in (w.index.names or []) else w
df = pd.concat([wi.rename('w'), S], axis=1).dropna(subset=['w'])
df[S.columns] = df[S.columns].fillna(0.0)

# Keep the shocks that are common enough to estimate.
keep = [c for c in S.columns if df[c].mean() > 0.02]
print(f"{len(keep)}_shock_types_with_{>2%}_incidence")

D = pd.get_dummies(df.index.get_level_values('t'), prefix='t',

```

```

                                drop_first=True, dtype=float)
D.index = df.index
X = pd.concat([df[keep], D], axis=1)
Z = pd.concat([df.w, X], axis=1)
Zd = Z - Z.groupby(level='i').transform('mean')      # household effects

b_s, V_s = clustered_ols(Zd.drop(columns='w'), Zd.w)
print(coef_table(b_s, V_s).to_string())

```

Read the shock coefficients, not the time dummies. Under full risk sharing every one of them is zero. Where they are not, that household bore the shock itself.

And then read what is missing. A drought that struck the whole country in a given wave contributes nothing to these estimates: the time dummy took it. Whatever this regression says about insurance, it says it only about the shocks that hit some households and not others in the same period.

6.10 Short panels and lagged outcomes

- **Short panels.** $y_{it} = \rho y_{i,t-1} + \alpha_i + \epsilon_{it}$ cannot be estimated by within: demeaning correlates the lagged dependent variable with the demeaned error (Nickell bias, order $1/T$). At $T = 3$ the bias in $\hat{\rho}$ can exceed 30%, downward. Arellano–Bond instruments in differences, and is weak precisely when ρ is near one.

6.11 Measurement and choice remain problems

- **Measurement error.** Recall error grows with the window and the item count, and it is not classical. Differencing removes signal and keeps noise, so fixed effects can make things worse, not better.
- **Endogeneity.** Land, labour supply, migration, schooling, adoption — all chosen. Fixed effects handle time-invariant selection only. Panel data does not solve endogeneity; it solves one special case of it.

6.12 Where it breaks

It is worth ending on this note because the temptation, having spent four sessions learning to handle these data carefully, is to over-claim with them. The instinct runs the wrong way: careful handling of the design makes the descriptive estimates trustworthy, and does nothing whatever for the causal ones.

The honest position is that household surveys are the best instrument we have for saying who is poor, where, by how much, and how that has changed — and that these are important questions which most of the profession's methodology is not aimed at. A well-constructed poverty profile with defensible standard errors is a more useful contribution than a badly-identified regression, and it is also harder to do. Session 6's purpose is to leave you willing to do the first and reluctant to do the second.

6.13 The capstone

Pick a country — `lsms_library.countries()` lists them — and produce, in about ninety minutes:

1. one paragraph on the design: frame, stages, strata, weights, structure
2. a consumption aggregate, or a defence of using the one provided
3. one weighted, correctly-clustered estimate you are willing to defend
4. one plot
5. a notebook that runs from a clean kernel, top to bottom

Five minutes each to present. The audience's job is to ask what the design lets you claim.

6.14 What counts as reproducible

- runs top to bottom in a fresh kernel, in order
- no manual steps, no "then I edited the CSV"
- data pulled by code, not sitting in a folder
- versions recorded — `lsms_library.__version__`, `pandas.__version__`
- someone else's machine, not just yours

This is not bureaucracy. It is the only way you will be able to answer a referee two years from now.

6.15 Setting up your capstone

```
import lsms_library as ll

# Everything the library knows about
ll.countries()

# Which tables exist where? 'coverage' is a country-by-table matrix.
# refresh='coverage' reads the declarations directly, so it does not need a
# prebuilt snapshot on disk.
cov = ll.coverage(refresh='coverage')
cov.head(20)

# Pick one and look before you leap
c = ll.Country('Uganda') # <- change me
print(c.waves)
print(c.data_scheme)
```

6.16 A reproducibility check

Put this at the top of the notebook you present.

```
import sys, platform
from importlib.metadata import version
import lsms_library, pandas, numpy, datamat

print(f"python_{sys.version.split()[0]}_{platform.system()}")
print(f"lsms_library_{lsms_library.__version__}")
print(f"CFEDemands_{version('CFEDemands')}")
print(f"pandas_{pandas.__version__}")
print(f"numpy_{numpy.__version__}")
print(f"datamat_{version('datamat')}")
```

If you have edited a country's configuration or scripts, the parquet cache will be stale, and there is no automatic staleness check:

```
# Force a rebuild for one country (slow; only when you have changed a confi
# !LSMS_NO_CACHE=1 python -c "import lsms_library as ll; ll.Country('Uganda
```

6.17 Exercises

The capstone is the exercise. Two suggestions if you finish early:

1. Take your estimate from (3) and compute it a second way — a different aggregate, a different scale, a different clustering level. Report the range. That range, not the standard error, is your real uncertainty.
2. Hand your notebook to the person next to you and have them run it from a clean kernel. Fix whatever breaks. That is the exercise.